INCH-POUND

MIL-M-28787/539
5 MAY 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

HIGH PERFORMANCE ADDRESS GENERATION MODULE, KEY CODE CWK

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic high performance address generation module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 4.700 failures per million operating hours for class I or 11.888 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CWK.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	5	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	5	------Module shall pass all remaining tests----------						
D1	Ibd	V	5	0.0	2.0	0.0	2.0	0.0	2.0	mA
D2	Ibd	V	5	0.0	2.1	0.0	2.1	0.0	2.1	mA
D3	Ibd	V	5	0.0	2.0	0.0	2.0	0.0	2.0	mA
D4	Ibd	V	5	0.0	1.1	0.0	1.1	0.0	1.1	mA
D5	Ibd	V	5	0.0	1.0	0.0	1.0	0.0	1.0	mA
D6	Ibd	V	5	0.0	1.1	0.0	1.1	0.0	1.1	mA
D7,D9	Ibd	V	5	0.0	1.0	0.0	1.0	0.0	1.0	mA
D8-D12	Ibd	V	5	0.0	2.1	0.0	2.1	0.0	2.1	mA
D13	Ibd	V	5	0.0	3.1	0.0	3.1	0.0	3.1	mA
D14	Ibd	V	5	0.0	1.0	0.0	1.0	0.0	1.0	mA
D15,D16	Ibd	V	5	0.0	2.3	0.0	2.3	0.0	2.3	mA
D17	Ibd	V	5	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	V	5	0.0	0.1	0.0	0.1	0.0	0.1	mA
D19	Ibd	V	5	0.0	1.5	0.0	1.5	0.0	1.5	mA
D20.0	Ibd	V	5	0.0	0.2	0.0	0.2	0.0	0.2	mA
D21	Ibd	V	5	0.0	2.0	0.0	2.0	0.0	2.0	mA
D22	Ibd	V	5	0.0	0.3	0.0	0.3	0.0	0.3	mA
D23	Ibd	V	5	0.0	2.1	0.0	2.1	0.0	2.1	mA
D24	Ibd	V	5	0.0	0.4	0.0	0.4	0.0	0.4	mA
D25	Ibd	V	5	0.0	0.6	0.0	0.6	0.0	0.6	mA
E1-E3	Ios	VI	5	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E4	Ios	VI	5	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E5	Ios	VI	5	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
E6	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E7-E10	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E11	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E12-E14	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E15	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E16-E18	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E19	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E20-E22	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E23	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E24-E26	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E27	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E28-E30	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E31	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E32-E34	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
E35	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E36,E37	Ios	VI	S	-325.0	-101.0	-325.0	-101.0	-325.0	-101.0	mA
E38-E43	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E44	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E45	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E46	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E47	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E48	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E49	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E50	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E51	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E52	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E53	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E54	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E55	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA

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				Min	Max	Min	Max	Min	Max	
E56	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E57	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E58	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E59	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E60-E63	Ios	VI	S	-326.0	-101.0	-326.0	-101.0	-326.0	-101.0	mA
E64-E66	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E67	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E68	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E69	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E70	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E71	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E72-E74	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E75-E90	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E91	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E92	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E93-E100	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
F1	Ilh	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F2	Ilh	VII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
F3	Ilh	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F4	Ilh	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F5,F6	Ilh	VII	X	-1.72	0.28	-1.72	0.28	-1.72	0.28	mA
F7-F12	Ilh	VII	X	-1.93	0.31	-1.93	0.31	-1.93	0.31	mA
F13	Ilh	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F14	Ilh	VII	X	-0.9	0.08	-0.9	0.08	-0.9	0.08	mA
F15-F22	Ilh	VII	X	-0.27	0.09	-0.27	0.09	-0.27	0.09	mA
F23,F24	Ilh	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F25-F28	Ilh	VII	X	-0.9	0.07	-0.9	0.07	-0.9	0.07	mA
F29	Ilh	VII	X	-1.15	0.10	-1.15	0.10	-1.15	0.10	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F30	Ilh	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F31,F32	Ilh	VII	X	-2.3	0.11	-2.3	0.11	-2.3	0.11	mA
F33	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F34	Ilh	VII	X	-16.8	0.42	-16.8	0.42	-16.8	0.42	mA
F35	Ilh	VII	X	-3.0	0.15	-3.0	0.15	-3.0	0.15	mA
F36-F43	Ilh	VII	X	-0.24	0.06	-0.24	0.06	-0.24	0.06	mA
F44	Ilh	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F45	Ilh	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F46	Ilh	VII	X	-0.8	0.08	-0.8	0.08	-0.8	0.08	mA
F47	Ilh	VII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
F48	Ilh	VII	X	-0.0	0.00	-0.0	0.00	-0.0	0.00	mA
F49	Ilh	VII	X	-6.0	0.12	-6.0	0.12	-6.0	0.12	mA
G1	Ill	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G2	Ill	VIII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
G3	Ill	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G4	Ill	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G5,G6	Ill	VIII	X	-1.72	0.28	-1.72	0.28	-1.72	0.28	mA
G7-G12	Ill	VIII	X	-1.93	0.31	-1.93	0.31	-1.93	0.31	mA
G13	Ill	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G14	Ill	VIII	X	-0.9	0.06	-0.9	0.06	-0.9	0.06	mA
G15-G22	Ill	VIII	X	-0.27	0.09	-0.27	0.09	-0.27	0.09	mA
G23,G24	Ill	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G25-G28	Ill	VIII	X	-0.9	0.07	-0.9	0.07	-0.9	0.07	mA
G29	Ill	VIII	X	-1.15	0.10	-1.15	0.10	-1.15	0.10	mA
G30	Ill	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G31,G32	Ill	VIII	X	-2.3	0.11	-2.3	0.11	-2.3	0.11	mA
G33	Ill	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G34	Ill	VIII	X	-16.8	0.42	-16.8	0.42	-16.8	0.42	mA
G35	Ill	VIII	X	-3.0	0.15	-3.0	0.15	-3.0	0.15	mA
G36-G43	Ill	VIII	X	-0.24	0.06	-0.24	0.06	-0.24	0.06	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ B ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
G44	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G45	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G46	Iil	VIII	X	-0.8	0.06	-0.8	0.06	-0.8	0.06	mA
G47	Iil	VIII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
G48	Iil	VIII	X	-0.8	0.08	-0.8	0.08	-0.8	0.08	mA
G49	Iil	VIII	X	-6.0	0.12	-6.0	0.12	-6.0	0.12	mA
H1-H100	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J100	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K17	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K18-K34	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1	Vio	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L2	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L3	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L4	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L5	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L6	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L7-L13	Vio	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L14-L20	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L21,L22	Vio	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L23	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L24	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L25	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	3.3	7.4	3.3	7.4	3.3	7.4	uF
M2	PSFC	XIII	S	2.3	4.4	2.3	4.4	2.3	4.4	uF

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
N	Icc	XIV	X	892	2139	892	2139	892	2139	mA
P1,P2	Functional	XV	X	----------		See table XV		----------		
Q1,Q2	Dynamic	XVI	S	----------		See table XVI		----------		
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI		----------		
R1-R3	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R4	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R5	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R6	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R7	C1/o	XVIII	Q	2	125	2	125	2	125	pF
R8	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R9	C1/o	XVIII	Q	8	100	8	100	8	100	pF
R10-R12	C1/o	XVIII	Q	6	75	6	75	6	75	pF
R13	C1/o	XVIII	Q	2	175	2	175	2	175	pF
R14	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R15	C1/o	XVIII	Q	4	150	4	150	4	150	pF
R16	C1/o	XVIII	Q	10	125	10	125	10	125	pF
R17	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R18	C1/o	XVIII	Q	10	175	10	175	10	175	pF
R19	C1/o	XVIII	Q	6	175	6	175	6	175	pF
R20	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R21	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R22	C1/o	XVIII	Q	4	150	4	150	4	150	pF
R23	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R24	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R25	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R26,R27	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R28-R33	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R34-R37	C1/o	XVIII	Q	4	75	4	75	4	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class T 4/ 0 ± 5°C 80 ± 5°C		Class H 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R38-R45	Ci/o	XVIII	0	2	75	2	75	2	75	pF
R46	Ci/o	XVIII	0	2	50	2	50	2	50	pF
R47	Ci/o	XVIII	0	2	75	2	75	2	75	pF
R48-R61	Ci/o	XVIII	0	6	100	6	100	6	100	pF
R62	Ci/o	XVIII	0	6	125	6	125	6	125	pF
T1, T2	PSTR	XIX	0	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/539-1 for class I and M28787/539-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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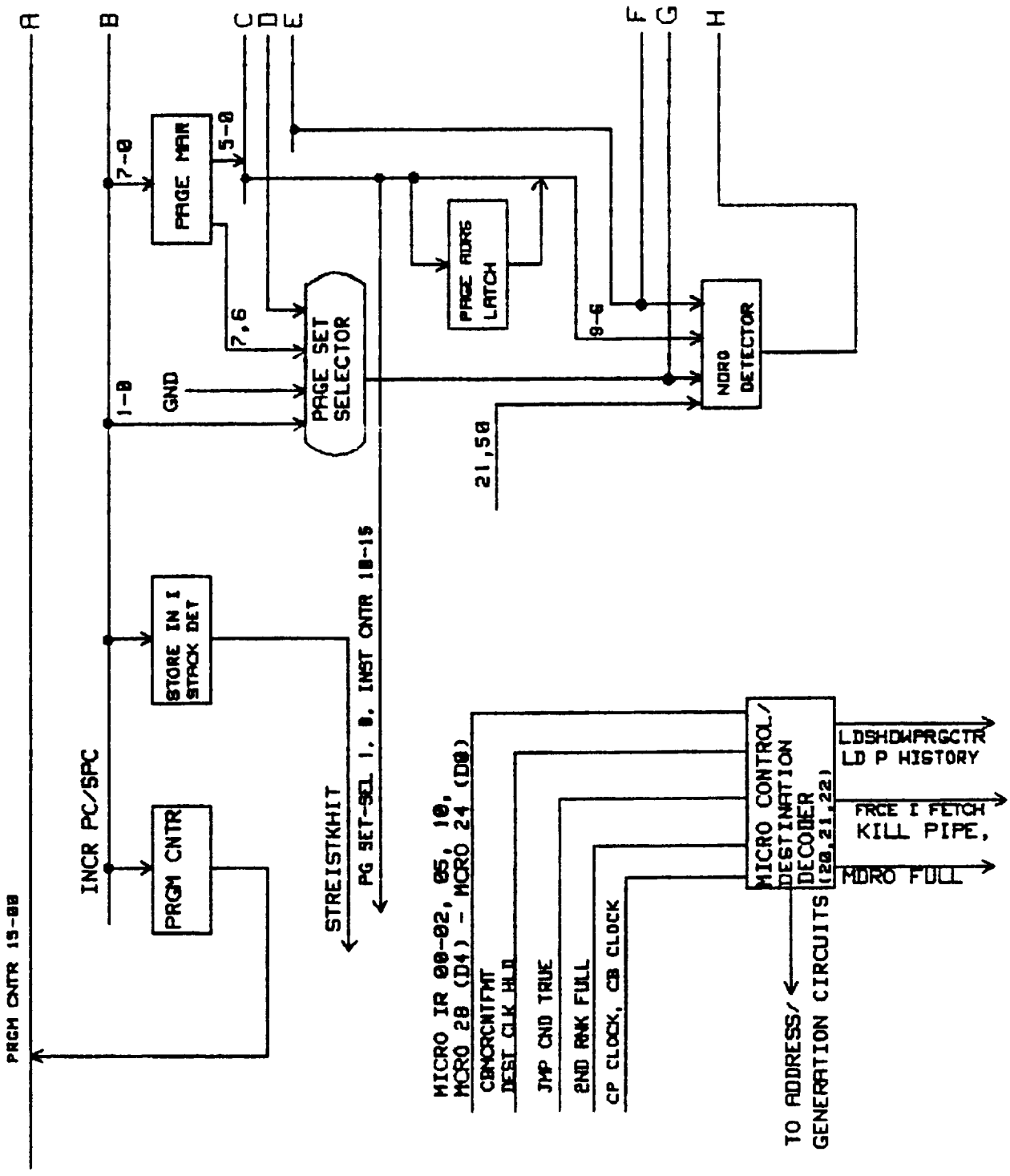


FIGURE 1. CWK functional block diagram.

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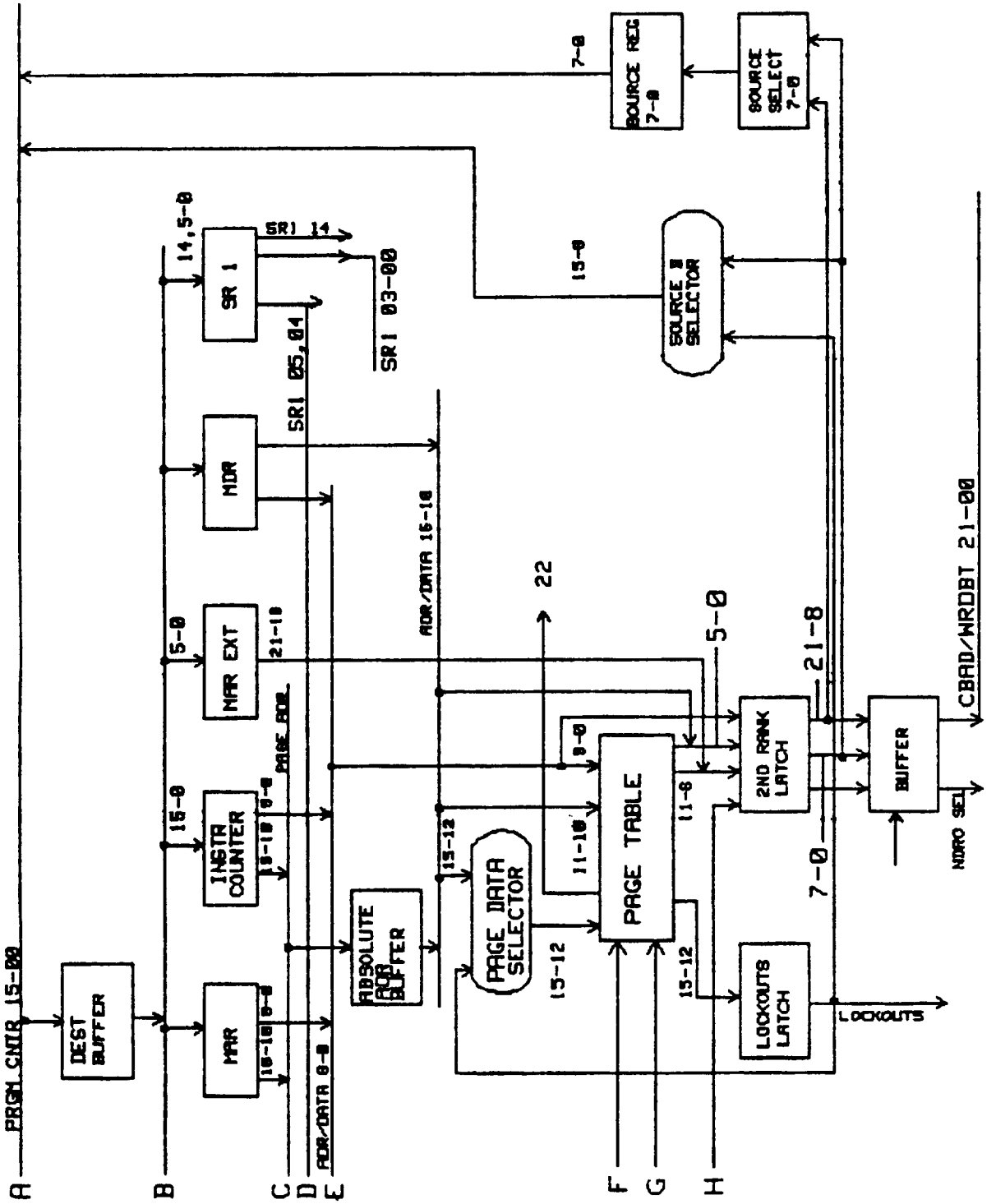


FIGURE 1. CWK functional block diagram - Continued.

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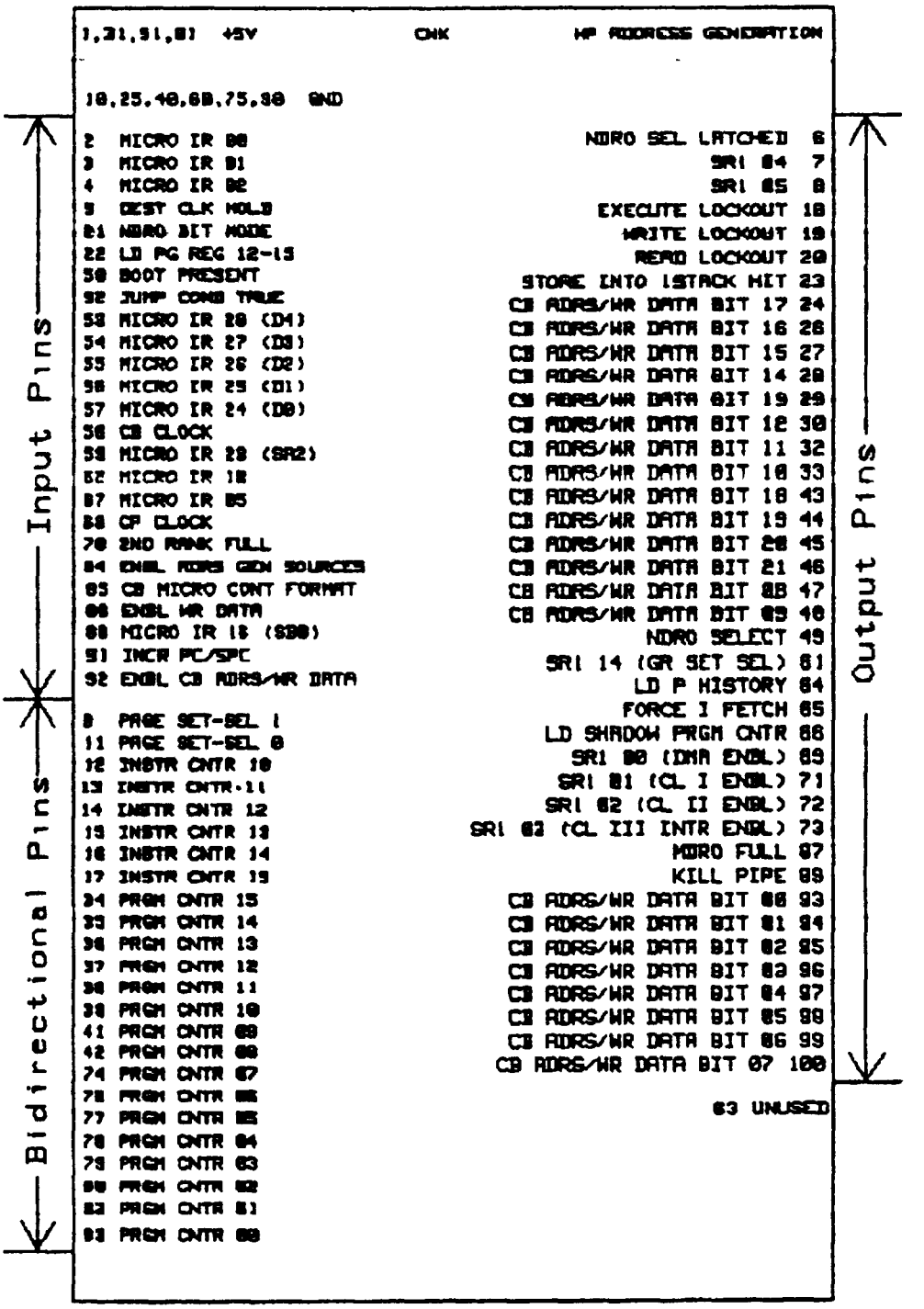


FIGURE 2. CWK pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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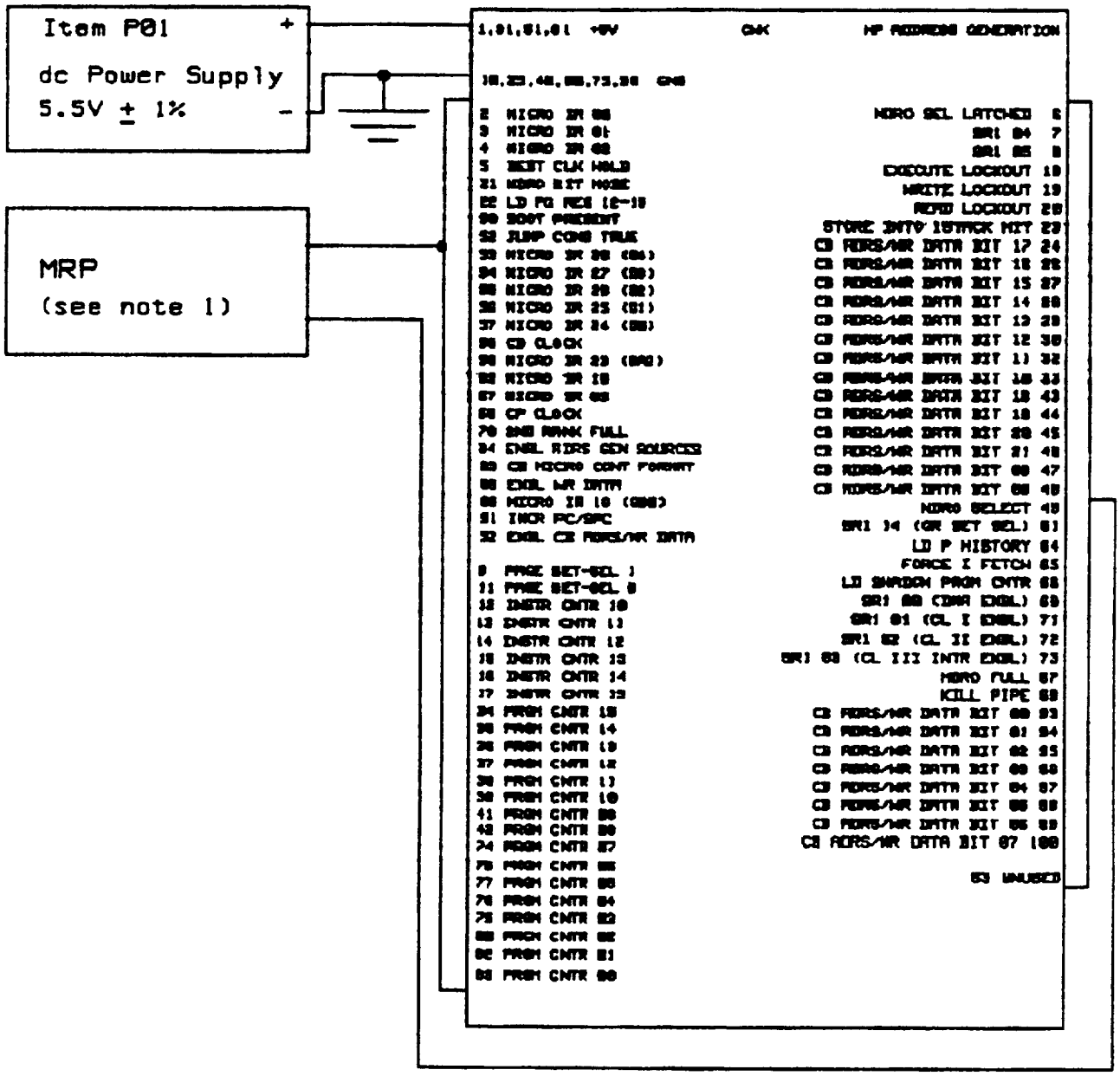
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5° C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. CWK operating temperature and component temperature test circuit.

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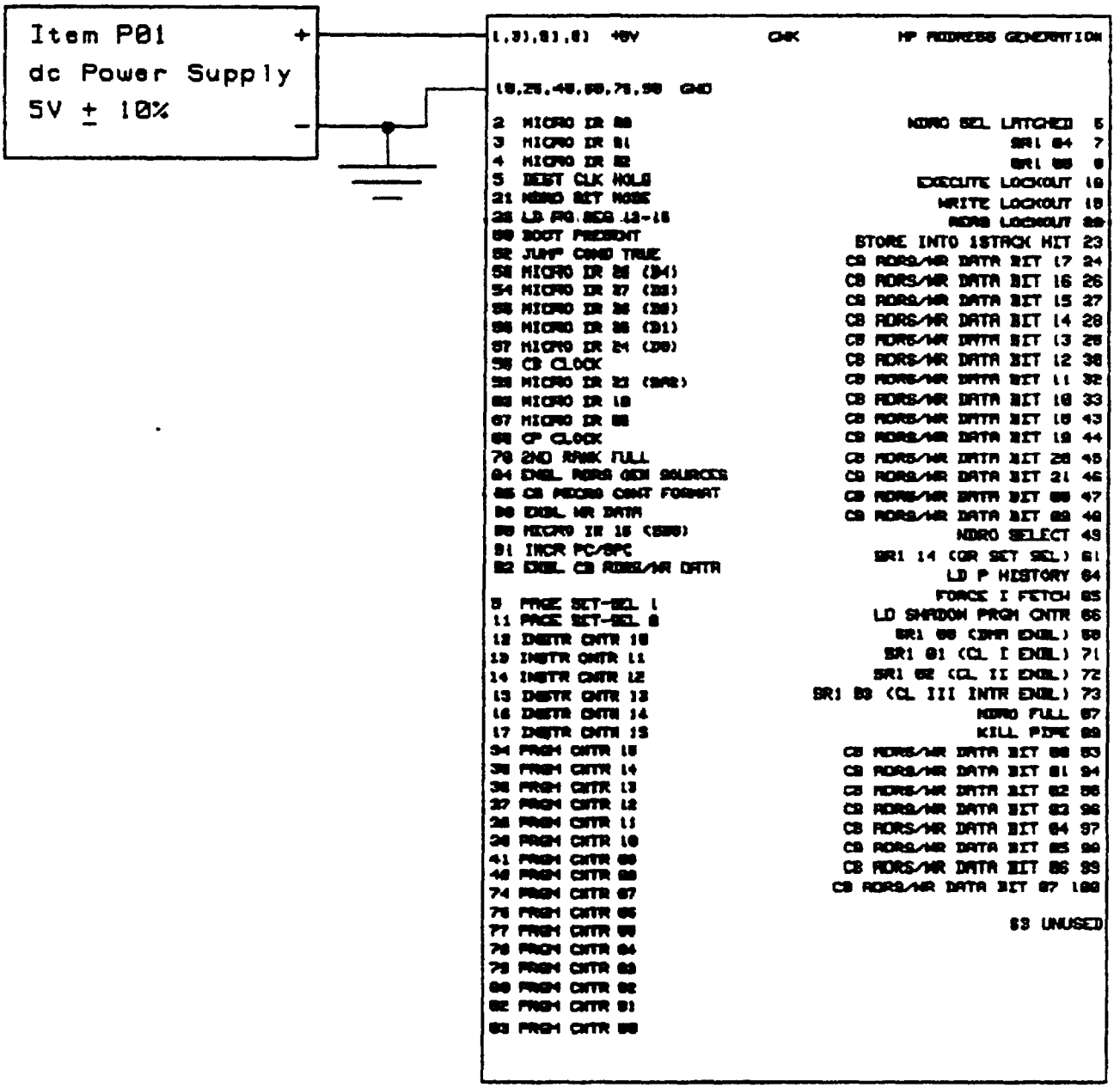
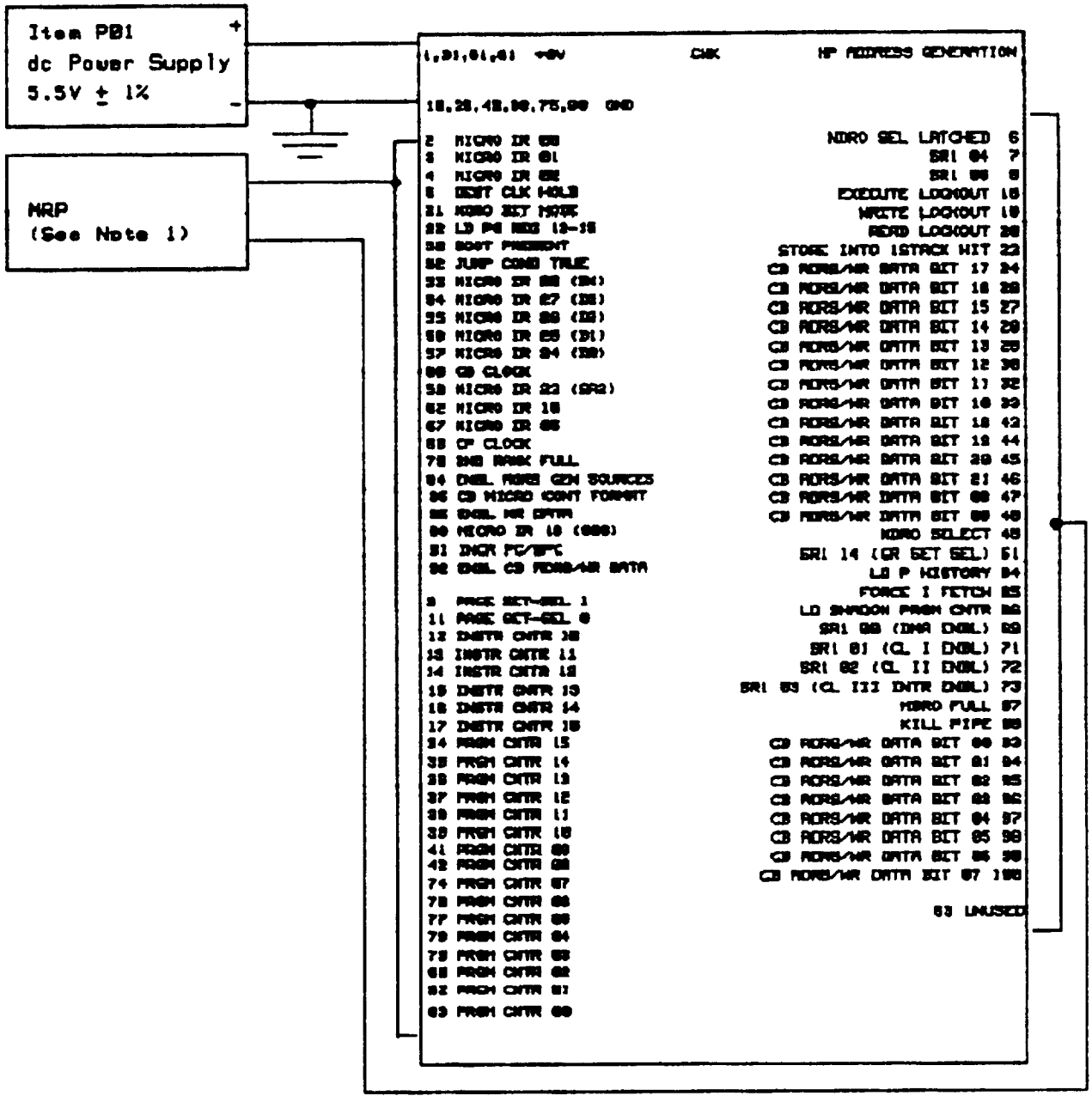


FIGURE 4. CWK vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. CWK life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 11.765 W (max); 8.34 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MICRO IR 00	-1.5	0.4	2.4	-0.5	0.05	2.0	75
MICRO IR 01	-1.2	0.4	2.4	-1.1	0.07	2.1	75
MICRO IR 02	-1.5	0.4	2.4	-0.5	0.05	2.0	75
DEST CLK HLD	-1.2	0.4	2.4	-2.6	0.07	1.1	100
NDRO BIT MDE	-1.5	0.4	2.4	-0.25	0.03	1.0	75
LDPGRG 12-15	-1.2	0.4	2.4	-0.9	0.06	1.1	100
BOOT PRESENT	-1.5	0.4	2.4	-0.25	0.03	1.0	125
JMP CND TRUE	-1.5	0.4	2.4	-0.25	0.03	1.0	75
MCRO 28 (D4)	-1.5	0.4	2.4	-0.9	0.07	2.1	100
MCRO 27 (D3)	-1.5	0.4	2.4	-0.9	0.07	2.1	75
MCRO 26 (D2)	-1.5	0.4	2.4	-0.9	0.07	2.1	75
MCRO 25 (D1)	-1.5	0.4	2.4	-0.9	0.07	2.1	75
MCRO 24 (D0)	-1.5	0.4	2.4	-1.15	0.10	3.1	175
CB CLOCK	-1.2	0.4	2.4	-2.0	0.05	1.0	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MCRO 23(SA2)	-1.2	0.4	2.4	-2.3	0.11	2.3	150
MICRO IR 10	-1.2	0.4	2.4	-2.3	0.11	2.3	125
MICRO IR 05	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CP CLOCK	-1.2	0.4	2.4	-16.6	0.42	8.1	175
2ND RNK FULL	-1.2	0.4	2.4	-3.8	0.15	1.5	175
ENBADRSGENSC	-1.2	0.4	2.4	-1.2	0.04	0.2	100
CBMCRCNTFMT	-1.5	0.4	2.4	-0.5	0.05	2.0	125
ENBL WR DATA	-1.5	0.4	2.4	-0.6	0.06	0.3	150
MCRO 16(SB0)	-1.2	0.4	2.4	-1.1	0.07	2.1	100
INCR PC/SPC	-1.5	0.4	2.4	-0.8	0.08	0.4	125
ENBCBAD/WRDA	-1.2	0.4	2.4	-6.0	0.12	0.6	150
PG SET-SEL 1	NA	0.4	2.4	-1.72	0.28	NA	NA
PG SET-SEL 0	NA	0.4	2.4	-1.72	0.28	NA	NA
INST CNTR 10	NA	0.4	2.4	-1.93	0.31	NA	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
INST CNTR 11	NA	0.4	2.4	-1.93	0.31	NA	NA
INST CNTR 12	NA	0.4	2.4	-1.93	0.31	NA	NA
INST CNTR 13	NA	0.4	2.4	-1.93	0.31	NA	NA
INST CNTR 14	NA	0.4	2.4	-1.93	0.31	NA	NA
INST CNTR 15	NA	0.4	2.4	-1.93	0.31	NA	NA
PRGM CNTR 15	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 14	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 13	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 12	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 11	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 10	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 09	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 08	NA	0.4	2.4	-0.27	0.09	NA	100
PRGM CNTR 07	NA	0.4	2.4	-0.24	0.06	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PRGM CNTR 06	NA	0.4	2.4	-0.24	0.06	NA	NA
PRGM CNTR 05	NA	0.4	2.4	-0.24	0.06	NA	100
PRGM CNTR 04	NA	0.4	2.4	-0.24	0.06	NA	100
PRGM CNTR 03	NA	0.4	2.4	-0.24	0.06	NA	100
PRGM CNTR 02	NA	0.4	2.4	-0.24	0.06	NA	100
PRGM CNTR 01	NA	0.4	2.4	-0.24	0.06	NA	100
PRGM CNTR 00	NA	0.4	2.4	-0.24	0.06	NA	125

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
NDRO SEL LTC	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
SR1 04	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
SR1 05	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
EXEC LCKOUT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
WRITE LCKOUT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
READ LCKOUT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
STREISTKHIT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CBAD/WRDBT17	2.4	0.5	-0.4	6.0	NA	NA	-101.0	75
CBAD/WRDBT16	2.4	0.5	-0.4	6.0	NA	NA	-101.0	75
CBAD/WRDBT15	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT14	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT13	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT12	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT11	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CBAD/WRDBT10	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT18	2.4	0.5	-0.4	6.0	NA	NA	-101.0	75
CBAD/WRDBT19	2.4	0.5	-0.4	6.0	NA	NA	-101.0	75
CBAD/WRDBT20	2.4	0.5	-0.4	6.0	NA	NA	-101.0	75
CBAD/WRDBT21	2.4	0.5	-0.4	6.0	NA	NA	-101.0	75
CBAD/WRDBT08	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT09	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
NDRO SELECT	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
SR1 14 (GSS)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
LD P HISTORY	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
FRCE I FETCH	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
LDSHDWPRGCTR	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
SR1 00 DMAEN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
SR1 01 CLIEN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SR1 02 CIIEN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SR1 03 CIIIE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MDRO FULL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
KILL PIPE	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CBAD/WRDBT00	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT01	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT02	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT03	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT04	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT05	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CBAD/WRDBT06	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	50
CBAD/WRDBT07	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
PG SET-SEL 1	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	NA
PG SET-SEL 0	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
INST CNTR 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
INST CNTR 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
INST CNTR 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
INST CNTR 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
INST CNTR 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
INST CNTR 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
PRGM CNTR 15	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 14	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 13	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 12	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 11	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 10	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 09	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100
PRGM CNTR 08	2.4	0.5	-0.16	1.6	NA	NA	-60.0 -30.0	100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
PRGM CNTR 07	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
PRGM CNTR 06	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
PRGM CNTR 05	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
PRGM CNTR 04	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
PRGM CNTR 03	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
PRGM CNTR 02	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
PRGM CNTR 01	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
PRGM CNTR 00	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP CLOCK to MDRO FULL	-	-	-	30	30	30	nS	12
2	tpd	CP CLOCK to SR1 00 DMAEN	-	-	-	40	40	40	nS	12
2	tpd	CP CLOCK to SR1 04	-	-	-	40	40	40	nS	12
2	tpd	CP CLOCK to SR1 05	-	-	-	40	40	40	nS	12
2	tpd	CP CLOCK to SR1 14 (GSS)	-	-	-	40	40	40	nS	12
3	tpd	CP CLOCK to SR1 01 CLIEN	-	-	-	45	45	45	nS	12
3	tpd	CP CLOCK to SR1 02 CIIEN	-	-	-	45	45	45	nS	12
3	tpd	CP CLOCK to SR1 03 CIIIE	-	-	-	45	45	45	nS	12
4	tpd	MICRO IR 00 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 00 to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 01 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 01 to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 02 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 02 to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 10 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 10 to KILL PIPE	-	-	-	55	55	55	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	MCRO 28 (D4) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 28 (D4) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 27 (D3) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 27 (D3) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 26 (D3) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 26 (D3) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 25 (D1) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 25 (D1) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 24 (D0) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 24 (D0) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	JMP CND TRUE to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	JMP CND TRUE to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	CBMCRCNTFMT to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	CBMCRCNTFMT to KILL PIPE	-	-	-	55	55	55	nS	12
5	tpd	MICRO IR 00 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	MICRO IR 00 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 01 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 01 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 02 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 02 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 10 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 10 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 28 (D4) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 28 (D4) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 27 (D3) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 27 (D3) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 26 (D2) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 26 (D2) to LD P HISTORY	-	-	-	60	60	60	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	MCRO 25 (D1) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 25 (D1) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 24 (D0) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 24 (D0) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	JMP CND TRUE to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	JMP CND TRUE to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	CBMCRCNTFMT to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	CBMCRCNTFMT to LD P HISTORY	-	-	-	60	60	60	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 15	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 14	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 13	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 12	-	-	-	80	80	80	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	ENBADRSGENSC to PRGM CNTR 11	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 10	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 09	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 08	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 07	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 06	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 05	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 04	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 03	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 02	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 01	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 00	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 15	-	-	-	80	80	80	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MCRO 16(SB0) to PRGM CNTR 14	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 13	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 12	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 11	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 10	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 09	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 08	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 07	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 06	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 05	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 04	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 03	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 02	-	-	-	80	80	80	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MCRO 16(SB0) to PRGM CNTR 01	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 00	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 15	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 14	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 13	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 12	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 11	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 10	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 09	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 08	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 07	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 06	-	-	-	80	80	80	nS	12
		MCRO 23(SA2) to PRGM CNTR 05								

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MCRO 23(SA2) to PRGM CNTR 04	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 03	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 02	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 01	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 00	-	-	-	80	80	80	nS	12
7	tpd	2ND RNK FULL to EXEC LCKOUT	-	-	-	30	30	30	nS	12
7	tpd	2ND RNK FULL to WRITE LCKOUT	-	-	-	30	30	30	nS	12
7	tpd	2ND RNK FULL to READ LCKOUT	-	-	-	30	30	30	nS	12
7	tpd	2ND RNK FULL to NDRO SEL LTC	-	-	-	30	30	30	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT17	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT16	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT15	-	-	-	20	20	20	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ENCBAD/WRDA to CBAD/WRDBT14	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT13	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT12	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT11	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT10	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT18	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT19	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT20	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT21	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT08	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT09	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT00	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT01	-	-	-	20	20	20	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ENCBAD/WRDA to CBAD/WRDBT02	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT03	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT04	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT05	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT06	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT07	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to NDRO SELECT	-	-	-	20	20	20	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT15	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT14	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT13	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT12	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT11	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT10	-	-	-	65	65	65	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	ENBL WR DATA to CBAD/WRDBT08	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT09	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT00	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT01	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT02	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT03	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT04	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT05	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT06	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT07	-	-	-	65	65	65	nS	12
10	tsu	MICRO IR 00 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MICRO IR 01 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MICRO IR 02 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MICRO IR 05 to CP CLOCK	100	100	100	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tsu	MICRO IR 10 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 28 (D4) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 27 (D3) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 26 (D2) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 25 (D1) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 24 (D0) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	JMP CND TRUE to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	CBMCRCNTFMT to CP CLOCK	100	100	100	-	-	-	nS	12
11	tsu	PRGM CNTR 15 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 14 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 13 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 12 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 11 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 10 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 09 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 08 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 07 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 06 to CP CLOCK	50	50	50	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	PRGM CNTR 05 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 04 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 03 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 02 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 01 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 00 to CP CLOCK	50	50	50	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 15	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 14	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 13	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 12	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 11	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 10	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 09	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 08	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 07	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 06	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 05	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 04	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 03	15	15	15	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	th	CP CLOCK to PRGM CNTR 02	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 01	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 00	15	15	15	-	-	-	nS	12
13	tsu	INCR PC/SPC to CP CLOCK	40	40	40	-	-	-	nS	12
14	tsu	2ND RNK FULL to CB CLOCK	15	15	15	-	-	-	nS	12
15	th	CB CLOCK to 2ND RNK FULL	5	5	5	-	-	-	nS	12
16	tsu	DEST CLK HLD to CP CLOCK	15	15	15	-	-	-	nS	12
17	th	CP CLOCK to DEST CLK HLD	5	5	5	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to PG SET-SEL 1	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to PG SET-SEL 0	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 10	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 11	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 12	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 13	15	15	15	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
18	tsu	LDPGRG 12-15 to INST CNTR 14	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 15	15	15	15	-	-	-	nS	12
19	th	PG SET-SEL 1 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	PG SET-SEL 0 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 10 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 11 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 12 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 13 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 14 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 15 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
20	tsu	NDRO BIT MDE to 2ND RNK FULL	60	60	60	-	-	-	nS	12
20	tsu	BOOT PRESENT to 2ND RNK FULL	60	60	60	-	-	-	nS	12

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6.4 Module function. The address generation (CWK) module contains the page table and associated registers that generate the memory address. A 22-bit absolute address is generated from a 16-bit relative address by the selected page register and transferred to the memory interface via the common bus. The Memory Address Register (MAR) holds the 16-bit relative address. MAR holds the relative address for operand read/write operations; the instruction counter holds the relative address of the next instruction to be fetched, and the program counter holds the relative address of the instruction currently being executed. The page table contains four sets of 64 page registers. Status register 1 bits 5 and 4 select a set. Each page register is used to address a page containing 1024 memory words. Relative address bits 9-0 are used to address a word within the page. Relative address bits 15-10 are used to address a page register that contains the fixed base address of a page within main memory. The 12-bit fixed address and the 10-bit relative address form the 22-bit absolute address. Each page register also contains a flag bit (bit 15) and three protect bits. The flag bit is set whenever data is written into any of the 1024 page locations. The execute, write, and read protect bits provide lockout protection against executing a macroinstruction from the page, writing data into the page, or reading data from the page. When the protect bit is set, and the corresponding memory operation is attempted, a class II protect fault interrupt is generated. The page registers must be loaded by a program using the B0 through B3 macroinstructions. On a master clear, the flag and protect bits are cleared and each page register is loaded with its own address, for example, page register 7 loaded with address 007. Therefore, in the initial condition, all memory references are made as though directly addressed. The memory data register holds the data written into a page register or the memory write data. The address source register holds the data read from a page register.

6.4.1 Address generation. The address generation module has a Program Counter (PC), Instruction (I) counter, Page (PG) MAR, and a store in I stack detector.

6.4.1.1 Memory address register. The 16-bit MAR holds the relative memory address for operand read/write operations and load/store page register operations. The register is loaded by a microinstruction with D = 0D or a format 3 microinstruction with bits 10, 2-0 = 9 or C. The register output is enabled by a format 3 microinstruction with Micro Control Instruction (MC) = 4-B and bits 10, 2-0 = 8-D. Also, the MAR lower byte output is enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = E (PG MAR output is also enabled).

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6.4.1.2 Memory address register extension. The 6-bit MAR extension holds destination bus bits 5-0 with the output inserted into address/data bus bits 21-16. This allows bypassing the page registers to address memory. The register is loaded by a microinstruction with D = 0E. The output is enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = A.

6.4.1.3 Memory Data Register (MDR). The 16-bit register holds memory write data. The register is loaded by a microinstruction with D = 0B. The register output is enabled by a bus grant signal or a format 3 microinstruction with MC = 4-B and bits 9-7 = 2.

6.4.1.4 Absolute address buffer. The 6-bit buffer holds MAR bits 15-10. The output is inserted into address/data bits 15-10 to form a part of an absolute memory address. This allows bypassing the page registers to address memory. The output is enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = A.

6.4.1.5 Page address register. The 6-bit page address register selects one of 64 page registers within a set. The register is loaded and enabled by a bus grant signal or a format 3 microinstruction with MC = 4-B and bits 9-7 = 2.

6.4.1.6 Page set selector. The page set selector determines what source is used to select one of the four page sets. The selector is enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 not equal to A or F, or by a microinstruction with D = 05. Selector inputs S1 and S0 select the source as follows:

S1	S0	
0	0	- Status register 1 bits 5 and 4
0	1	- Destination bus bits 1 and 0
1	0	- Ground (to select page set 0)
1	1	- Page MAR bits 7 and 6

The value of S1 and S0 is determined by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 except when a force I fetch operation is performed.

(a) S1 and S0 = 00 when bits 10, 0-3, 5, 8, 9, D or a force I fetch operation (D = 05) is performed.

(b) S1 and S0 = 01 when bits 10, 2-0 = B.

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(c) S1 and S0 = 01 when bits 10, 2-0 = 4, 6, 7 or C.

(d) S1 and S0 = 11 when bits 10, 2-0 = E.

6.4.1.7 Page table. The page table contains 256 16-bit page registers. The page table is divided into four sets of 64 registers. A set is selected by the outputs from the page set selector. A page register within a set is addressed by page address register bits 5-0. A page register is loaded by a format 3 microinstruction with MC = 4 through B and bits 10, 2-0 = E. Bits 15-12 are also reloaded on memory references via the page data selector. Page table output bits 11-0 are placed in address/data bus bits 21-10 to form the absolute memory address along with MAR bits 9-0. Output bits 11-0 are enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 not equal to A or F, or by a microinstruction with D = 05 (force I fetch). Output bits 15-12 are always enabled.

6.4.1.8 Page data selector. The 4-bit page data selector loads the addressed page register bits 15-12. The selector loads address/data bus bits 15-12 when a page register is loaded by a format 3 microinstruction with MC = 4 through B and bits 10, 2-0 = E. The selector reloads the addressed page register with bits 14-12 (lockouts) and inserts a one in bit 15 when the page register is used for a memory reference.

6.4.1.9 Page MAR. The 8-bit page MAR holds the page register address for load/store page register operations. The register is loaded by a microinstruction with D = 0D or a format 3 microinstruction with bits 10, 2-0 = 9 or C. The register output is enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = E. Output bits 7 and 6 are used to select a page set and bits 5-0 address a register within the set.

6.4.1.10 Instruction counter. The 16-bit instruction counter holds the relative memory address of the next instruction to be fetched. The instruction counter is loaded by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = 2-7 or a microinstruction with D = 05 and jump condition true. The counter is incremented by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = 07 or a microinstruction with D = 05. The counter is incremented by one when bits 10, 2-0 = 4 or 6, all memory writes, and memory reads from odd addresses. The counter is incremented by two when bits 10, 2-0 = 0-3, 5, or 7, or on memory reads from even addresses. The counter output is enabled by a format 3 microinstruction with MC = 4-B and bits 10, 2-0 = 0-7 or by a microinstruction with D = 05 (force I fetch).

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6.4.1.11 Program counter. The 16-bit program counter holds the relative memory address of the instruction currently being executed. The program counter is loaded by a microinstruction with D = 06. The counter is incremented by a format 3 microinstruction with MC = 4-B and bit 11 = 1. The counter output is enabled by a microinstruction with SB = 1C and page set 1 is not selected.

6.4.1.12 Store in I stack detector. The store in I stack detector determines if an operand is being stored at a memory location where an instruction has been fetched from but has not yet been executed (instruction in I stack or pipe). Once detected, a micro interrupt is generated to prevent executing the obsolete instruction. The detection process is part of the operand store micro code. The PC (holds address of current instruction) is subtracted from the operand address with the result placed on the source B/Destination bus. The result is sampled by the detector. For a normal operation (D field bit 0 = 0), if the result is 5 or less, the store into stack hit is generated causing a micro interrupt. For multiple store operation (D field bit 0 = 1), if the result is 19 (decimal) or less, the hit is generated.

6.4.1.13 Status Register 1 (SR 1). The 7-bit register holds SR 1 bits 14, 5-0. The bits duplicate the corresponding SR 1 bits on the emulation control 1 module. The register is loaded by a microinstruction with D = 01 (same as SR 1 on emulation control module). The output is always enabled.

6.4.1.14 Nondestructive Readout (NDRO) detector. The NDRO detector determines if the relative address is a NDRO memory address. It detects boot strap memory addresses and NDRO Built-In-Test (BIT) addresses. Bootstrap memory is located in the 0000 - 003F and 00C0 - 013F address ranges. BIT is located in the 0400-07FF address range. The detector provides a NDRO reference signal if the address is within the NDRO address range and the bootstrap memory/Real Time Clock (RTC) module is present, page set 0 is selected, the MAR extension register output is not enabled, and the NDRO BIT mode input is low (BIT address range only).

6.4.1.15 2nd rank and lockouts latches. The 2nd rank latch holds the 22-bit absolute memory address and NDRO reference signal or memory write data. The lockouts latch holds page register bits 15-12 (page modify, execute lockout, write lockout, and read lockout).

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The latches are loaded by a bus grant signal or a format 3 microinstruction with MC = 4 through B, bits 10, 2-0 = 0-E, and bits 9-7 = 0. The latches outputs are always enabled. The absolute memory address or memory write data is placed on the common bus via the buffer.

6.4.1.16 Address source circuits. The address source circuits consist of a source bus multiplexer (bits 15-8), source register 7-0, and a source detector 7-0. The address source circuits select the address source to place on the source B/Destination bus. The address placed on the bus depends on the last microcontrol operation (format 3 microinstruction with MC = 4-B). The address bits are placed on the bus by the source circuits when a microinstruction with SB = 1D or 1C (and page set was selected by Rm+1 bits 1 and 0) is executed for each microcontrol operation. The program counter is sourced when SB = 1C for the other page sets selections.

6.4.1.17 Micro control/destination decoder. The micro control circuits interpret microinstruction bits 19, 2-0 to provide the control signals for the paging and bus operations. The destination decoder translates the microinstruction D-field to provide register enables.

6.4.2 Description of I/O pins.

Signal name	Pin number	I/O	Description
MICRO IR 00	2	I	MICRO IR 00
MICRO IR 01	3	I	MICRO IR 01
MICRO IR 02	4	I	MICRO IR 02
DEST CLK HLD	5	I	DEST CLK HOLD
NDRO BIT MDE	21	I	NDRO BIT MODE
LDPGRG 12-15	22	I	LD PG REG 12-15
BOOT PRESENT	50	I	BOOT PRESENT
JMP CND TRUE	52	I	JUMP COND TRUE

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MCRO 28 (D4)	53	I	MICRO IR 28 (D4)
MCRO 27 (D3)	54	I	MICRO IR 27 (D3)
MCRO 26 (D2)	55	I	MICRO IR 26 (D2)
MCRO 25 (D1)	56	I	MICRO IR 25 (D1)
MCRO 24 (D0)	57	I	MICRO IR 24 (D0)
CB CLOCK	58	I	CB CLOCK
MCRO 23(SA2)	59	I	MICRO IR 23 (SA2)
MICRO IR 10	62	I	MICRO IR 10
MICRO IR 05	67	I	MICRO IR 05
CP CLOCK	68	I	CP CLOCK
2ND RNK FULL	70	I	2ND RANK FULL
ENBADRSGENSC	84	I	ENABLE ADRS GEN SOURCES
CBMCRCNTFMT	85	I	CB MICRO CONT FORMAT
ENBL WR DATA	86	I	ENBL WR DATA
MCRO 16(SB0)	88	I	MICRO IR 16 (SB0)
INCR PC/SPC	91	I	INCR PC/SPC
ENBCBAD/WRDA	92	I	ENBL CB ADRS/WR DATA
NDRO SEL LTC	6	O	NDRO SEL LATCHED
SR1 04	7	O	SR1 04
SR1 05	8	O	SR1 05
EXEC LCKOUT	18	O	EXECUTE LOCKOUT

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
WRITE LCKOUT	19	O	WRITE LOCKOUT
READ LCKOUT	20	O	READ LOCKOUT
STREISTKHIT	23	O	STORE INTO ISTACK HIT
CBAD/WRDBT17	24	O	CB ADRS/WR DATA BIT 17
CBAD/WRDBT16	26	O	CB ADRS/WR DATA BIT 16
CBAD/WRDBT15	27	O	CB ADRS/WR DATA BIT 15
CBAD/WRDBT14	28	O	CB ADRS/WR DATA BIT 14
CBAD/WRDBT13	29	O	CB ADRS/WR DATA BIT 13
CBAD/WRDBT12	30	O	CB ADRS/WR DATA BIT 12
CBAD/WRDBT11	32	O	CB ADRS/WR DATA BIT 11
CBAD/WRDBT10	33	O	CB ADRS/WR DATA BIT 10
CBAD/WRDBT18	43	O	CB ADRS/WR DATA BIT 18
CBAD/WRDBT19	44	O	CB ADRS/WR DATA BIT 19
CBAD/WRDBT20	45	O	CB ADRS/WR DATA BIT 20
CBAD/WRDBT21	46	O	CB ADRS/WR DATA BIT 21
CBAD/WRDBT08	47	O	CB ADRS/WR DATA BIT 08
CBAD/WRDBT09	48	O	CB ADRS/WR DATA BIT 09
NDRO SELECT	49	O	NDRO SELECT
SR1 14 (GSS)	61	O	SR1 14 (GR SET SEL)
LD P HISTORY	64	O	LD P HISTORY
FRCE I FETCH	65	O	FORCE I FETCH

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
LDSHDWPRGCTR	66	O	LOAD SHADOW PRGM CNTR
SR1 00 DMAEN	69	O	SR1 00 (DMA ENBL)
SR1 01 CLIEN	71	O	SR1 01 (CL I ENBL)
SR1 02 CIIEN	72	O	SR1 02 (CL II ENBL)
SR1 03 CIIIE	73	O	SR1 03 (CL III INTR ENBL)
MDRO FULL	87	O	MDRO FULL
KILL PIPE	89	O	KILL PIPE
CBAD/WRDBT00	93	O	CB ADRS/WR DATA BIT 00
CBAD/WRDBT01	94	O	CB ADRS/WR DATA BIT 01
CBAD/WRDBT02	95	O	CB ADRS/WR DATA BIT 02
CBAD/WRDBT03	96	O	CB ADRS/WR DATA BIT 03
CBAD/WRDBT04	97	O	CB ADRS/WR DATA BIT 04
CBAD/WRDBT05	98	O	CB ADRS/WR DATA BIT 05
CBAD/WRDBT06	99	O	CB ADRS/WR DATA BIT 06
CBAD/WRDBT07	100	O	CB ADRS/WR DATA BIT 07
PG SET-SEL 1	9	I/O	PAGE SET-SEL 1
PG SET-SEL 0	11	I/O	PAGE SET-SEL 0
INST CNTR 10	12	I/O	INSTR CNTR 10
INST CNTR 11	13	I/O	INSTR CNTR 11
INST CNTR 12	14	I/O	INSTR CNTR 12

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6.4.2 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
INST CNTR 13	15	I/O	INSTR CNTR 13
INST CNTR 14	16	I/O	INSTR CNTR 14
INST CNTR 15	17	I/O	INSTR CNTR 15
PRGM CNTR 15	34	I/O	PRGM CNTR 15
PRGM CNTR 14	35	I/O	PRGM CNTR 14
PRGM CNTR 13	36	I/O	PRGM CNTR 13
PRGM CNTR 12	37	I/O	PRGM CNTR 12
PRGM CNTR 11	38	I/O	PRGM CNTR 11
PRGM CNTR 10	39	I/O	PRGM CNTR 10
PRGM CNTR 09	41	I/O	PRGM CNTR 09
PRGM CNTR 08	42	I/O	PRGM CNTR 08
PRGM CNTR 07	74	I/O	PRGM CNTR 07
PRGM CNTR 06	76	I/O	PRGM CNTR 06
PRGM CNTR 05	77	I/O	PRGM CNTR 05
PRGM CNTR 04	78	I/O	PRGM CNTR 04
PRGM CNTR 03	79	I/O	PRGM CNTR 03
PRGM CNTR 02	80	I/O	PRGM CNTR 02
PRGM CNTR 01	82	I/O	PRGM CNTR 01
PRGM CNTR 00	83	I/O	PRGM CNTR 00

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy-SH

(Project 5963-N715)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,63	Frame	10	INF	MOhm	1-8
A2	Isolation, unused pin to used pins	63	C	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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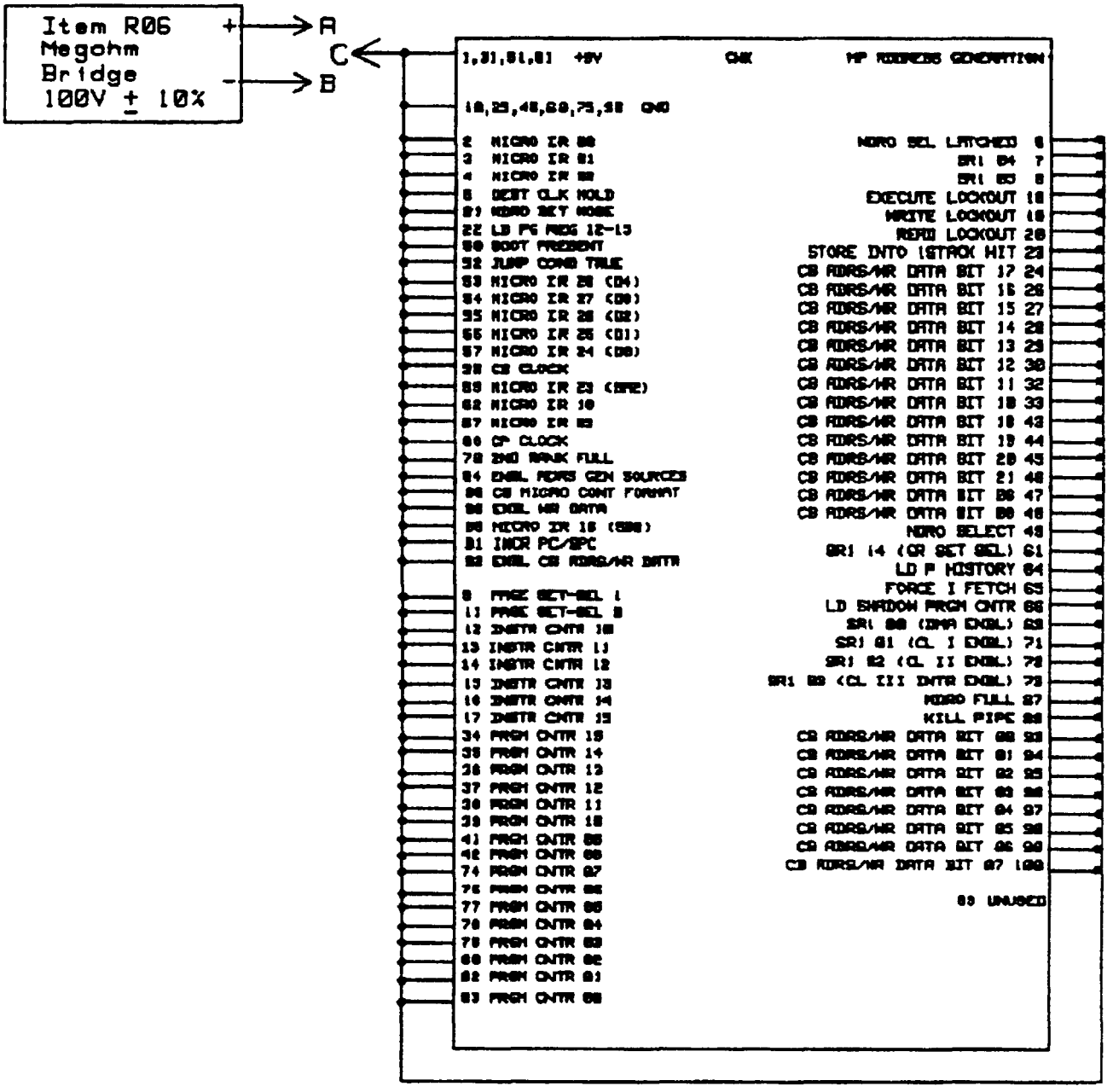


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, ground	10	25	0	1	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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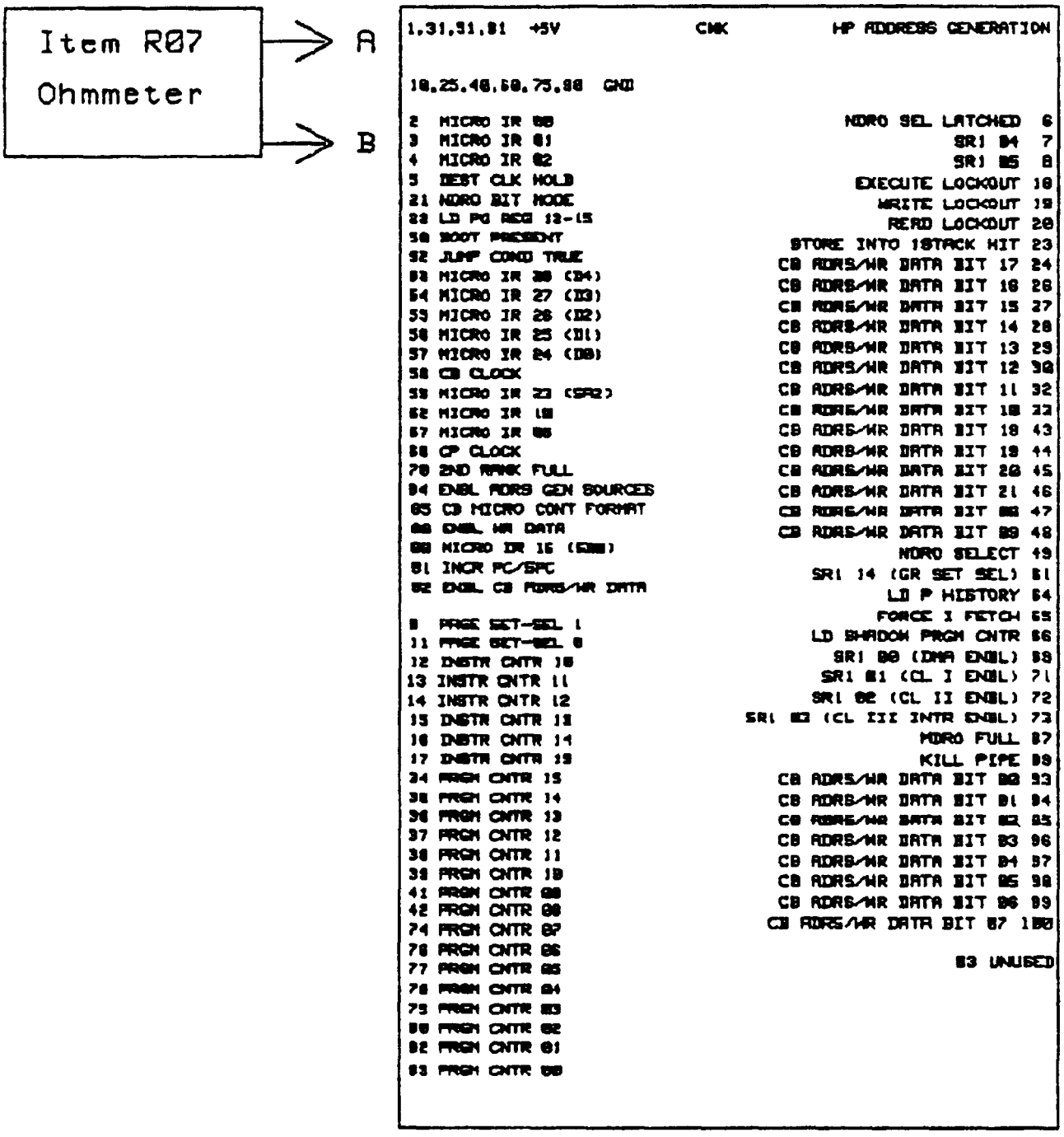


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.85 A	± 39 mA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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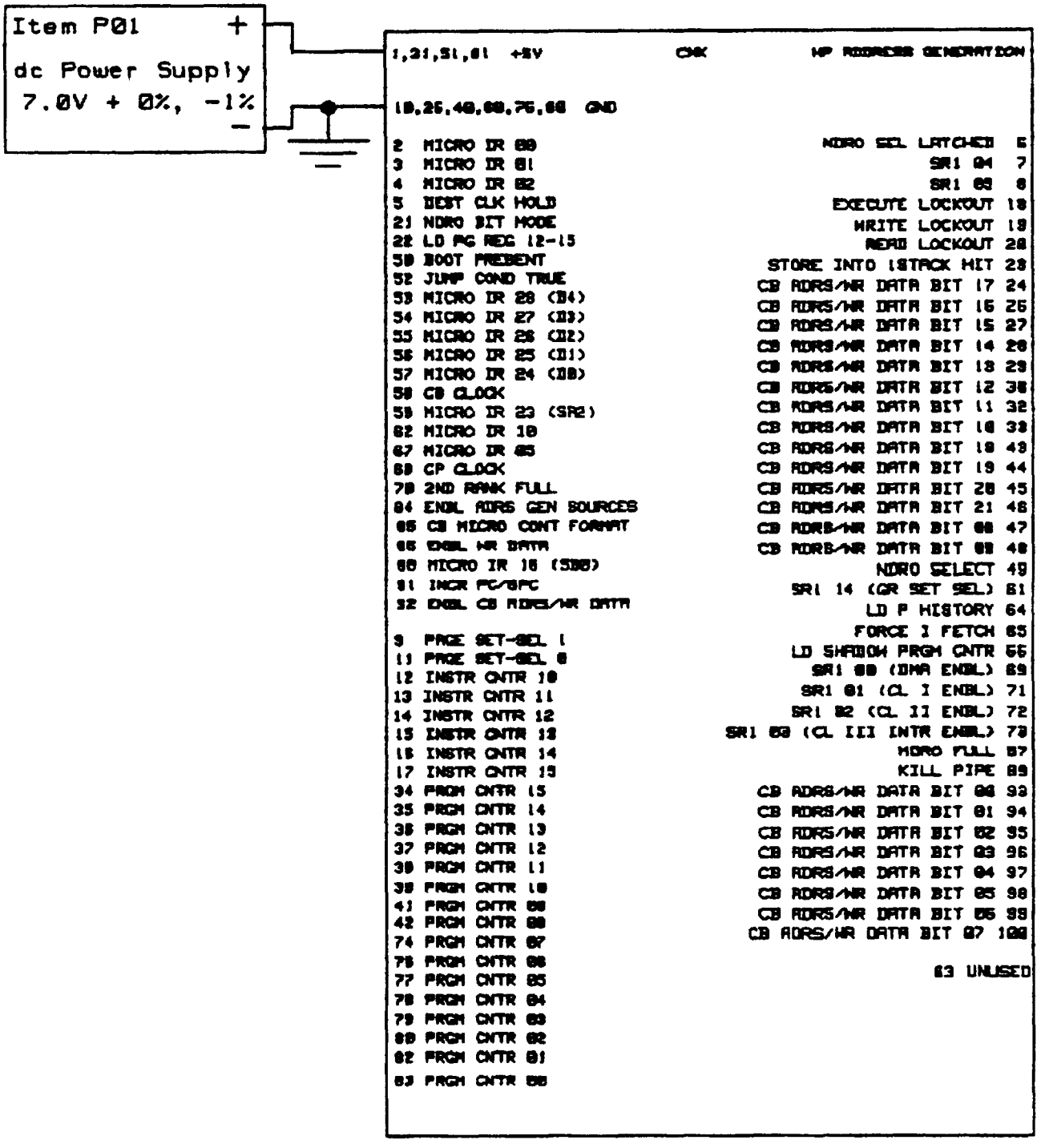


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	MICRO IR 00	2	5.5	0.0	2.0	mA	1-3
D2	MICRO IR 01	3	5.5	0.0	2.1	mA	1-3
D3	MICRO IR 02	4	5.5	0.0	2.0	mA	1-3
D4	DEST CLK HLD	5	5.5	0.0	1.1	mA	1-3
D5	NDRO BIT MDE	21	5.5	0.0	1.0	mA	1-3
D6	LDPGRG 12-15	22	5.5	0.0	1.1	mA	1-3
D7	BOOT PRESENT	50	5.5	0.0	1.0	mA	1-3
D8	JMP CND TRUE	52	5.5	0.0	1.0	mA	1-3
D9	MCRO 28 (D4)	53	5.5	0.0	2.1	mA	1-3
D10	MCRO 27 (D3)	54					
D11	MCRO 26 (D2)	55					
D12	MCRO 25 (D1)	56	5.5	0.0	2.1	mA	1-3
D13	MCRO 24 (D0)	57	5.5	0.0	3.1	mA	1-3
D14	CB CLOCK	58	5.5	0.0	1.0	mA	1-3
D15	MCRO 23(SA2)	59	5.5	0.0	2.3	mA	1-3
D16	MICRO IR 10	62	5.5	0.0	2.3	mA	1-3
D17	MICRO IR 05	67	5.5	0.0	0.1	mA	1-3
D18	CP CLOCK	68	5.5	0.0	8.1	mA	1-3
D19	2ND RNK FULL	70	5.5	0.0	1.5	mA	1-3
D20	ENBADRSGENSC	84	5.5	0.0	0.2	mA	1-3
D21	CBMCRCNTFMT	85	5.5	0.0	2.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	ENBL WR DATA	86	5.5	0.0	0.3	mA	1-3
D23	MCRO 16(SB0)	88	5.5	0.0	2.1	mA	1-3
D24	INCR PC/SPC	91	5.5	0.0	0.4	mA	1-3
D25	ENBCBAD/WRDA	92	5.5	0.0	0.6	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D25
Power supply current limit	3.20 A	NA	D1-D25
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D25
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D25
low, (VL)	0.0 V	± 100 mV	D1-D25
Input current measurement	NA	± 100 uA	D1
	NA	± 105 uA	D2
	NA	± 100 uA	D3
	NA	± 50 uA	D4
	NA	± 50 uA	D5
	NA	± 55 uA	D6

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NOTES: - Continued.

Condition	Value	Accuracy	Test
Input current measurement	NA	± 50 uA	D7,D8
	NA	± 105 uA	D9-D12
	NA	± 155 uA	D13
	NA	± 50 uA	D14
	NA	± 115 uA	D15,D16
	NA	± 5 uA	D17
	NA	± 405 uA	D18
	NA	± 75 uA	D19
	NA	± 10 uA	D20
	NA	± 100 uA	D21
	NA	± 15 uA	D22
	NA	± 105 uA	D23
	NA	± 20 uA	D24
	NA	± 30 uA	D25

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	NDRO SEL LTC	6	0.0	-260.0	-30.0	mA	1-4
E2	SR1 04	7					
E3	SR1 05	8	0.0	-260.0	-30.0	mA	1-4
E4	PG SET-SEL 1	9	0.0	-150.0	-60.0	mA	1-4
E5	PG SET-SEL 1	9	0.0	-260.0	-30.0	mA	1-4
E6	PG SET-SEL 0	11	0.0	-150.0	-60.0	mA	1-4
E7	PG SET-SEL 0	11	0.0	-260.0	-30.0	mA	1-4
E8	INST CNTR 10	12					
E9	INST CNTR 10	12					
E10	INST CNTR 10	12	0.0	-260.0	-30.0	mA	1-4
E11	INST CNTR 10	12	0.0	-220.0	-30.0	mA	1-4
E12	INST CNTR 11	13	0.0	-260.0	-30.0	mA	1-4
E13	INST CNTR 11	13					
E14	INST CNTR 11	13	0.0	-260.0	-30.0	mA	1-4
E15	INST CNTR 11	13	0.0	-220.0	-30.0	mA	1-4
E16	INST CNTR 12	14	0.0	-260.0	-30.0	mA	1-4
E17	INST CNTR 12	14					
E18	INST CNTR 12	14	0.0	-260.0	-30.0	mA	1-4
E19	INST CNTR 12	14	0.0	-220.0	-30.0	mA	1-4
E20	INST CNTR 13	15	0.0	-260.0	-30.0	mA	1-4
E21	INST CNTR 13	15	0.0	-260.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	INST CNTR 13	15	0.0	-260.0	-30.0	mA	1-4
E23	INST CNTR 13	15	0.0	-220.0	-30.0	mA	1-4
E24	INST CNTR 14	16	0.0	-260.0	-30.0	mA	1-4
E25	INST CNTR 14	16					
E26	INST CNTR 14	16	0.0	-260.0	-30.0	mA	1-4
E27	INST CNTR 14	16	0.0	-220.0	-30.0	mA	1-4
E28	INST CNTR 15	17	0.0	-260.0	-30.0	mA	1-4
E29	INST CNTR 15	17					
E30	INST CNTR 15	17	0.0	-260.0	-30.0	mA	1-4
E31	INST CNTR 15	17	0.0	-220.0	-30.0	mA	1-4
E32	EXEC LCKOUT	18	0.0	-260.0	-30.0	mA	1-4
E33	WRITE LCKOUT	19					
E34	READ LCKOUT	20	0.0	-260.0	-30.0	mA	1-4
E35	STREISTKHIT	23	0.0	-130.0	-30.0	mA	1-4
E36	CBAD/WRDBT17	24	0.0	-326.0	-101.0	mA	1-4
E37	CBAD/WRDBT16	26	0.0	-326.0	-101.0	mA	1-4
E38	CBAD/WRDBT15	27	0.0	-325.0	-100.0	mA	1-4
E39	CBAD/WRDBT14	28					
E40	CBAD/WRDBT13	29					
E41	CBAD/WRDBT12	30					
E42	CBAD/WRDBT11	32	0.0	-325.0	-100.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	CBAD/WRDBT10	33	0.0	-325.0	-100.0	mA	1-4
E44	PRGM CNTR 15	34	0.0	-150.0	-60.0	mA	1-4
E45	PRGM CNTR 15	34	0.0	-220.0	-30.0	mA	1-4
E46	PRGM CNTR 14	35	0.0	-150.0	-60.0	mA	1-4
E47	PRGM CNTR 14	35	0.0	-220.0	-30.0	mA	1-4
E48	PRGM CNTR 13	36	0.0	-150.0	-60.0	mA	1-4
E49	PRGM CNTR 13	36	0.0	-220.0	-30.0	mA	1-4
E50	PRGM CNTR 12	37	0.0	-150.0	-60.0	mA	1-4
E51	PRGM CNTR 12	37	0.0	-220.0	-30.0	mA	1-4
E52	PRGM CNTR 11	38	0.0	-150.0	-60.0	mA	1-4
E53	PRGM CNTR 11	38	0.0	-220.0	-30.0	mA	1-4
E54	PRGM CNTR 10	39	0.0	-150.0	-60.0	mA	1-4
E55	PRGM CNTR 10	39	0.0	-220.0	-30.0	mA	1-4
E56	PRGM CNTR 09	41	0.0	-150.0	-60.0	mA	1-4
E57	PRGM CNTR 09	41	0.0	-220.0	-30.0	mA	1-4
E58	PRGM CNTR 08	42	0.0	-150.0	-60.0	mA	1-4
E59	PRGM CNTR 08	42	0.0	-220.0	-30.0	mA	1-4
E60	CBAD/WRDBT18	43	0.0	-326.0	-101.0	mA	1-4
E61	CBAD/WRDBT19	44					
E62	CBAD/WRDBT20	45					
E63	CBAD/WRDBT21	46	0.0	-326.0	-101.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E64	CBAD/WRDBT08	47	0.0	-325.0	-100.0	mA	1-4
E65	CBAD/WRDBT09	48					
E66	NDRO SELECT	49	0.0	-325.0	-100.0	mA	1-4
E67	SR1 14 (GSS)	61	0.0	-260.0	-30.0	mA	1-4
E68	LD P HISTORY	64	0.0	-100.0	-40.0	mA	1-4
E69	FRCE I FETCH	65	0.0	-130.0	-30.0	mA	1-4
E70	LDSHDWPRGCTR	66	0.0	-100.0	-40.0	mA	1-4
E71	SR1 00 DMAEN	69	0.0	-260.0	-30.0	mA	1-4
E72	SR1 01 CLIEN	71	0.0	-150.0	-60.0	mA	1-4
E73	SR1 02 CIIEN	72					
E74	SR1 03 CIIIE	73	0.0	-150.0	-60.0	mA	1-4
E75	PRGM CNTR 07	74	0.0	-220.0	-30.0	mA	1-4
E76	PRGM CNTR 07	74					
E77	PRGM CNTR 06	76					
E78	PRGM CNTR 06	76					
E79	PRGM CNTR 05	77					
E80	PRGM CNTR 05	77					
E81	PRGM CNTR 04	78					
E82	PRGM CNTR 04	78					
E83	PRGM CNTR 03	79					
E84	PRGM CNTR 03	79	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E85	PRGM CNTR 02	80	0.0	-220.0	-30.0	mA	1-4
E86	PRGM CNTR 02	80					
E87	PRGM CNTR 01	82					
E88	PRGM CNTR 01	82					
E89	PRGM CNTR 00	83					
E90	PRGM CNTR 00	83	0.0	-220.0	-30.0	mA	1-4
E91	MDRO FULL	87	0.0	-150.0	-60.0	mA	1-4
E92	KILL PIPE	89	0.0	-130.0	-30.0	mA	1-4
E93	CBAD/WRDBT00	93	0.0	-325.0	-100.0	mA	1-4
E94	CBAD/WRDBT01	94					
E95	CBAD/WRDBT02	95					
E96	CBAD/WRDBT03	96					
E97	CBAD/WRDBT04	97					
E98	CBAD/WRDBT05	98					
E99	CBAD/WRDBT06	99					
E100	CBAD/WRDBT07	100	0.0	-325.0	-100.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E100
Power supply current limit	3.20 A	NA	E1-E100
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E100
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E100
low, (VL)	0.0 V	± 100 mV	E1-E100
Output current measurement	NA	± 11.5 mA	E1-E3
	NA	± 4.50 mA	E4
	NA	± 11.5 mA	E5
	NA	± 4.50 mA	E6
	NA	± 11.5 mA	E7-E10
	NA	± 9.50 mA	E11
	NA	± 11.5 mA	E12-E14
	NA	± 9.50 mA	E15
	NA	± 11.5 mA	E16-E18
	NA	± 9.50 mA	E19
	NA	± 11.5 mA	E20-E22
	NA	± 9.50 mA	E23
	NA	± 11.5 mA	E24-E26
	NA	± 9.50 mA	E27
	NA	± 11.5 mA	E28-E30
	NA	± 9.50 mA	E31
	NA	± 11.5 mA	E32-E34
	NA	± 5.00 mA	E35
	NA	± 11.3 mA	E36-E43
	NA	± 4.50 mA	E44
	NA	± 9.50 mA	E45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 4.50 mA	E46
	NA	± 9.50 mA	E47
	NA	± 4.50 mA	E48
	NA	± 9.50 mA	E49
	NA	± 4.50 mA	E50
	NA	± 9.50 mA	E51
	NA	± 4.50 mA	E52
	NA	± 9.50 mA	E53
	NA	± 4.50 mA	E54
	NA	± 9.50 mA	E55
	NA	± 4.50 mA	E56
	NA	± 9.50 mA	E57
	NA	± 4.50 mA	E58
	NA	± 9.50 mA	E59
	NA	± 11.3 mA	E60-E66
	NA	± 11.5 mA	E67
	NA	± 3.00 mA	E68
	NA	± 5.00 mA	E69
	NA	± 3.00 mA	E70
	NA	± 11.5 mA	E71
	NA	± 4.50 mA	E72-E74
	NA	± 9.50 mA	E75-E90
	NA	± 4.50 mA	E91
	NA	± 5.00 mA	E92
	NA	± 11.3 mA	E93-E100

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	MICRO IR 00	2	2.4	-0.5	0.05	mA	1-3
F2	MICRO IR 01	3	2.4	-1.1	0.07	mA	1-3
F3	MICRO IR 02	4	2.4	-0.5	0.05	mA	1-3
F4	DEST CLK HLD	5	2.4	-2.6	0.07	mA	1-3
F5	PG SET-SEL 1	9	2.4	-1.72	0.28	mA	1-3
F6	PG SET-SEL 0	11	2.4	-1.72	0.28	mA	1-3
F7	INST CNTR 10	12	2.4	-1.93	0.31	mA	1-3
F8	INST CNTR 11	13					
F9	INST CNTR 12	14					
F10	INST CNTR 13	15					
F11	INST CNTR 14	16					
F12	INST CNTR 15	17	2.4	-1.93	0.31	mA	1-3
F13	NDRO BIT MDE	21	2.4	-0.25	0.03	mA	1-3
F14	LDPGRG 12-15	22	2.4	-0.9	0.06	mA	1-3
F15	PRGM CNTR 15	34	2.4	-0.27	0.09	mA	1-3
F16	PRGM CNTR 14	35					
F17	PRGM CNTR 13	36					
F18	PRGM CNTR 12	37					
F19	PRGM CNTR 11	38					
F20	PRGM CNTR 10	39					
F21	PRGM CNTR 09	41	2.4	-0.27	0.09	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
F22	PRGM CNTR 08	42	2.4	-0.27	0.09	mA	1-3
F23	BOOT PRESENT	50	2.4	-0.25	0.03	mA	1-3
F24	JMP CND TRUE	52	2.4	-0.25	0.03	mA	1-3
F25	MCRO 28 (D4)	53	2.4	-0.9	0.07	mA	1-3
F26	MCRO 27 (D3)	54					
F27	MCRO 26 (D2)	55					
F28	MCRO 25 (D1)	56	2.4	-0.9	0.07	mA	1-3
F29	MCRO 24 (D0)	57	2.4	-1.15	0.10	mA	1-3
F30	CB CLOCK	58	2.4	-2.0	0.05	mA	1-3
F31	MCRO 23(SA2)	59	2.4	-2.3	0.11	mA	1-3
F32	MICRO IR 10	62	2.4	-2.3	0.11	mA	1-3
F33	MICRO IR 05	67	2.4	-0.6	0.02	mA	1-3
F34	CP CLOCK	68	2.4	-16.6	0.42	mA	1-3
F35	2ND RNK FULL	70	2.4	-3.8	0.15	mA	1-3
F36	PRGM CNTR 07	74	2.4	-0.24	0.06	mA	1-3
F37	PRGM CNTR 06	76					
F38	PRGM CNTR 05	77					
F39	PRGM CNTR 04	78					
F40	PRGM CNTR 03	79					
F41	PRGM CNTR 02	80					
F42	PRGM CNTR 01	82	2.4	-0.24	0.06	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F43	PRGM CNTR 00	83	2.4	-0.24	0.06	mA	1-3
F44	ENBADRSGENSC	84	2.4	-1.2	0.04	mA	1-3
F45	CBMCRCNTFMT	85	2.4	-0.5	0.05	mA	1-3
F46	ENBL WR DATA	86	2.4	-0.6	0.06	mA	1-3
F47	MCRO 16(SB0)	88	2.4	-1.1	0.07	mA	1-3
F48	INCR PC/SPC	91	2.4	-0.8	0.08	mA	1-3
F49	ENBCBAD/WRDA	92	2.4	-6.0	0.12	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F49
Power supply current limit	3.20 A	NA	F1-F49
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F49
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F49
low, (VL)	0.0 V	± 100 mV	F1-F49

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 27.5 uA	F1
	NA	± 58.5 uA	F2
	NA	± 27.5 uA	F3
	NA	± 134.0 uA	F4
	NA	± 100.0 uA	F5,F6
	NA	± 112.0 uA	F7-F12
	NA	± 14.0 uA	F13
	NA	± 48.0 uA	F14
	NA	± 18.0 uA	F15-F22
	NA	± 14.0 uA	F23,F24
	NA	± 48.5 uA	F25-F28
	NA	± 62.5 uA	F29
	NA	± 103.0 uA	F30
	NA	± 121.0 uA	F31,F32
	NA	± 31.0 uA	F33
	NA	± 85.0 uA	F34
	NA	± 198.0 uA	F35
	NA	± 15.0 uA	F36-F43
	NA	± 62.0 uA	F44
	NA	± 27.5 uA	F45
	NA	± 33.0 uA	F46
	NA	± 58.5 uA	F47
	NA	± 44.0 uA	F48
	NA	± 306.0 uA	F49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	MICRO IR 00	2	0.4	-0.5	0.05	mA	1-3
G2	MICRO IR 01	3	0.4	-1.1	0.07	mA	1-3
G3	MICRO IR 02	4	0.4	-0.5	0.05	mA	1-3
G4	DEST CLK HLD	5	0.4	-2.6	0.07	mA	1-3
G5	PG SET-SEL 1	9	0.4	-1.72	0.28	mA	1-3
G6	PG SET-SEL 0	11	0.4	-1.72	0.28	mA	1-3
G7	INST CNTR 10	12	0.4	-1.93	0.31	mA	1-3
G8	INST CNTR 11	13					
G9	INST CNTR 12	14					
G10	INST CNTR 13	15					
G11	INST CNTR 14	16					
G12	INST CNTR 15	17	0.4	-1.93	0.31	mA	1-3
G13	NDRO BIT MDE	21	0.4	-0.25	0.03	mA	1-3
G14	LDPGRG 12-15	22	0.4	-0.9	0.06	mA	1-3
G15	PRGM CNTR 15	34	0.4	-0.27	0.09	mA	1-3
G16	PRGM CNTR 14	35					
G17	PRGM CNTR 13	36					
G18	PRGM CNTR 12	37					
G19	PRGM CNTR 11	38					
G20	PRGM CNTR 10	39					
G21	PRGM CNTR 09	41	0.4	-0.27	0.09	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	PRGM CNTR 08	42	0.4	-0.27	0.09	mA	1-3
G23	BOOT PRESENT	50	0.4	-0.25	0.03	mA	1-3
G24	JMP CND TRUE	52	0.4	-0.25	0.03	mA	1-3
G25	MCRO 28 (D4)	53	0.4	-0.9	0.07	mA	1-3
G26	MCRO 27 (D3)	54					
G27	MCRO 26 (D2)	55					
G28	MCRO 25 (D1)	56	0.4	-0.9	0.07	mA	1-3
G29	MCRO 24 (D0)	57	0.4	-1.15	0.10	mA	1-3
G30	CB CLOCK	58	0.4	-2.0	0.05	mA	1-3
G31	MCRO 23(SA2)	59	0.4	-2.3	0.11	mA	1-3
G32	MICRO IR 10	62	0.4	-2.3	0.11	mA	1-3
G33	MICRO IR 05	67	0.4	-0.6	0.02	mA	1-3
G34	CP CLOCK	68	0.4	-16.6	0.42	mA	1-3
G35	2ND RNK FULL	70	0.4	-3.8	0.15	mA	1-3
G36	PRGM CNTR 07	74	0.4	-0.24	0.06	mA	1-3
G37	PRGM CNTR 06	76					
G38	PRGM CNTR 05	77					
G39	PRGM CNTR 04	78					
G40	PRGM CNTR 03	79					
G41	PRGM CNTR 02	80					
G42	PRGM CNTR 01	82	0.4	-0.24	0.06	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	PRGM CNTR 00	83	0.4	-0.24	0.06	mA	1-3
G44	ENBADRSGENSC	84	0.4	-1.2	0.04	mA	1-3
G45	CBMCRCNTFMT	85	0.4	-0.5	0.05	mA	1-3
G46	ENBL WR DATA	86	0.4	-0.6	0.06	mA	1-3
G47	MCRO 16(SB0)	88	0.4	-1.1	0.07	mA	1-3
G48	INCR PC/SPC	91	0.4	-0.8	0.08	mA	1-3
G49	ENBCBAD/WRDA	92	0.4	-6.0	0.12	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G49
Power supply current limit	3.20 A	NA	G1-G49
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G49
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G49
low, (VL)	0.0 V	± 100 mV	G1-G49

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 27.5 uA	G1
	NA	± 58.5 uA	G2
	NA	± 27.5 uA	G3
	NA	± 134.0 uA	G4
	NA	± 100.0 uA	G5,G6
	NA	± 112.0 uA	G7-G12
	NA	± 14.0 uA	G13
	NA	± 48.0 uA	G14
	NA	± 18.0 uA	G15-G22
	NA	± 14.0 uA	G23,G24
	NA	± 48.5 uA	G25-G28
	NA	± 62.5 uA	G29
	NA	± 103.0 uA	G30
	NA	± 121.0 uA	G31,G32
	NA	± 31.0 uA	G33
	NA	± 851.0 uA	G34
	NA	± 198.0 uA	G35
	NA	± 15.0 uA	G36-G43
	NA	± 62.0 uA	G44
	NA	± 27.5 uA	G45
	NA	± 33.0 uA	G46
	NA	± 59.0 uA	G47
	NA	± 44.0 uA	G48
	NA	± 306.0 uA	G49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	NDRO SEL LTC	6	-0.16	2.4	4.5	V	1-4
H2	SR1 04	7					
H3	SR1 05	8					
H4	PG SET-SEL 1	9					
H5	PG SET-SEL 1	9					
H6	PG SET-SEL 0	11					
H7	PG SET-SEL 0	11					
H8	INST CNTR 10	12					
H9	INST CNTR 10	12					
H10	INST CNTR 10	12					
H11	INST CNTR 10	12					
H12	INST CNTR 11	13					
H13	INST CNTR 11	13					
H14	INST CNTR 11	13					
H15	INST CNTR 11	13					
H16	INST CNTR 12	14					
H17	INST CNTR 12	14					
H18	INST CNTR 12	14					
H19	INST CNTR 12	14					
H20	INST CNTR 13	15					
H21	INST CNTR 13	15	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	INST CNTR 13	15	-0.16	2.4	4.5	V	1-4
H23	INST CNTR 13	15					
H24	INST CNTR 14	16					
H25	INST CNTR 14	16					
H26	INST CNTR 14	16					
H27	INST CNTR 14	16					
H28	INST CNTR 15	17					
H29	INST CNTR 15	17					
H30	INST CNTR 15	17					
H31	INST CNTR 15	17					
H32	EXEC LCKOUT	18					
H33	WRITE LCKOUT	19					
H34	READ LCKOUT	20					
H35	STREISTKHIT	23	-0.16	2.4	4.5	V	1-4
H36	CBAD/WRDBT17	24	-0.4	2.4	4.5	V	1-4
H37	CBAD/WRDBT16	26					
H38	CBAD/WRDBT15	27					
H39	CBAD/WRDBT14	28					
H40	CBAD/WRDBT13	29					
H41	CBAD/WRDBT12	30					
H42	CBAD/WRDBT11	32	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H43	CBAD/WRDBT10	33	-0.4	2.4	4.5	V	1-4
H44	PRGM CNTR 15	34	-0.16	2.4	4.5	V	1-4
H45	PRGM CNTR 15	34					
H46	PRGM CNTR 14	35					
H47	PRGM CNTR 14	35					
H48	PRGM CNTR 13	36					
H49	PRGM CNTR 13	36					
H50	PRGM CNTR 12	37					
H51	PRGM CNTR 12	37					
H52	PRGM CNTR 11	38					
H53	PRGM CNTR 11	38					
H54	PRGM CNTR 10	39					
H55	PRGM CNTR 10	39					
H56	PRGM CNTR 09	41					
H57	PRGM CNTR 09	41					
H58	PRGM CNTR 08	42					
H59	PRGM CNTR 08	42	-0.16	2.4	4.5	V	1-4
H60	CBAD/WRDBT18	43	-0.4	2.4	4.5	V	1-4
H61	CBAD/WRDBT19	44					
H62	CBAD/WRDBT20	45					
H63	CBAD/WRDBT21	46	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H64	CBAD/WRDBT08	47	-0.4	2.4	4.5	V	1-4
H65	CBAD/WRDBT09	48					
H66	NDRO SELECT	49	-0.4	2.4	4.5	V	1-4
H67	SR1 14 (GSS)	61	-0.16	2.4	4.5	V	1-4
H68	LD P HISTORY	64	-0.4	2.4	4.5	V	1-4
H69	FRCE I FETCH	65	-0.16	2.4	4.5	V	1-4
H70	LDSHDWPRGCTR	66	-0.4	2.4	4.5	V	1-4
H71	SR1 00 DMAEN	69	-0.16	2.4	4.5	V	1-4
H72	SR1 01 CLIEN	71	-0.4	2.4	4.5	V	1-4
H73	SR1 02 CIIEN	72					
H74	SR1 03 CIIIE	73	-0.4	2.4	4.5	V	1-4
H75	PRGM CNTR 07	74	-0.16	2.4	4.5	V	1-4
H76	PRGM CNTR 07	74					
H77	PRGM CNTR 06	76					
H78	PRGM CNTR 06	76					
H79	PRGM CNTR 05	77					
H80	PRGM CNTR 05	77					
H81	PRGM CNTR 04	78					
H82	PRGM CNTR 04	78					
H83	PRGM CNTR 03	79					
H84	PRGM CNTR 03	79	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H85	PRGM CNTR 02	80	-0.16	2.4	4.5	V	1-4
H86	PRGM CNTR 02	80					
H87	PRGM CNTR 01	82					
H88	PRGM CNTR 01	82					
H89	PRGM CNTR 00	83					
H90	PRGM CNTR 00	83	-0.16	2.4	4.5	V	1-4
H91	MDRO FULL	87	-0.4	2.4	4.5	V	1-4
H92	KILL PIPE	89	-0.16	2.4	4.5	V	1-4
H93	CBAD/WRDBT00	93	-0.4	2.4	4.5	V	1-4
H94	CBAD/WRDBT01	94					
H95	CBAD/WRDBT02	95					
H96	CBAD/WRDBT03	96					
H97	CBAD/WRDBT04	97					
H98	CBAD/WRDBT05	98					
H99	CBAD/WRDBT06	99					
H100	CBAD/WRDBT07	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H100
Power supply current limit	3.20 A	NA	H1-H100
Forcing current, (Ioh)	-0.16 mA	± 16 uA	H1-H35
	-0.40 mA	± 40 uA	H36-H43
	-0.16 mA	± 16 uA	H44-H59
	-0.40 mA	± 40 uA	H60-H66
	-0.16 mA	± 16 uA	H67
	-0.40 mA	± 40 uA	H68
	-0.16 mA	± 16 uA	H69
	-0.40 mA	± 40 uA	H70
	-0.16 mA	± 16 uA	H71
	-0.40 mA	± 40 uA	H72-H74
	-0.16 mA	± 16 uA	H75-H90
	-0.40 mA	± 40 uA	H91
	-0.16 mA	± 16 uA	H92
	-0.40 mA	± 40 uA	H93-H100
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H100
low, (VL)	0.0 V	± 100 mV	H1-H100
Output voltage measurement	NA	± 105 mV	H1-H100

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

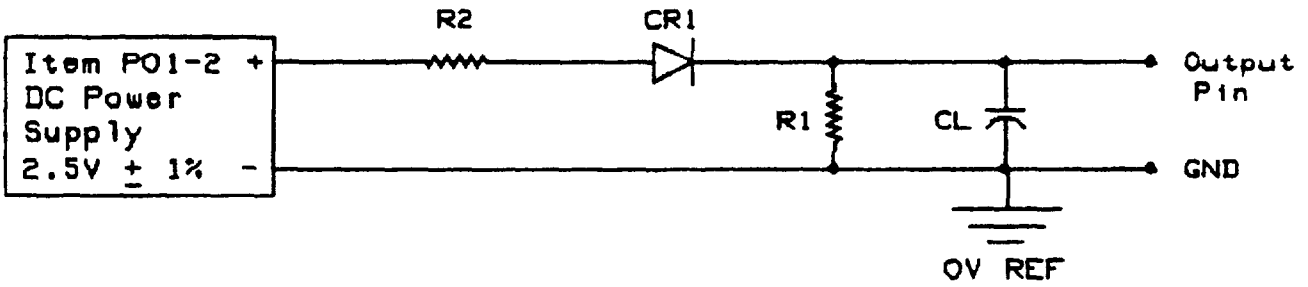
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R2 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R1 for each output load shall be as specified in the table below.

Pin number	R1 1/ (ohms)	Iol (mA)	Vol (volts)
6,7,8,9,11,12,13, 14,15,16,17,18,19, 20,23,34,35,36,37, 38,39,41,42,61,65, 69,74,76,77,78,79, 80,82,83,89	16000	1.6	0.5
24,26,27,28,29,30, 32,33,43,44,45,46, 47,48,49,64,66,71, 72,73,87,93,94,95, 96,97,98,99,100	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	NDRO SEL LTC	6	1.6	0.0	0.5	V	1-4
J2	SR1 04	7					
J3	SR1 05	8					
J4	PG SET-SEL 1	9					
J5	PG SET-SEL 1	9					
J6	PG SET-SEL 0	11					
J7	PG SET-SEL 0	11					
J8	INST CNTR 10	12					
J9	INST CNTR 10	12					
J10	INST CNTR 10	12					
J11	INST CNTR 10	12					
J12	INST CNTR 11	13					
J13	INST CNTR 11	13					
J14	INST CNTR 11	13					
J15	INST CNTR 11	13					
J16	INST CNTR 12	14					
J17	INST CNTR 12	14					
J18	INST CNTR 12	14					
J19	INST CNTR 12	14					
J20	INST CNTR 13	15					
J21	INST CNTR 13	15	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	INST CNTR 13	15	1.6	0.0	0.5	V	1-4
J23	INST CNTR 13	15					
J24	INST CNTR 14	16					
J25	INST CNTR 14	16					
J26	INST CNTR 14	16					
J27	INST CNTR 14	16					
J28	INST CNTR 15	17					
J29	INST CNTR 15	17					
J30	INST CNTR 15	17					
J31	INST CNTR 15	17					
J32	EXEC LCKOUT	18					
J33	WRITE LCKOUT	19					
J34	READ LCKOUT	20					
J35	STREISTKHIT	23	1.6	0.0	0.5	V	1-4
J36	CBAD/WRDBT17	24	6.0	0.0	0.5	V	1-4
J37	CBAD/WRDBT16	26					
J38	CBAD/WRDBT15	27					
J39	CBAD/WRDBT14	28					
J40	CBAD/WRDBT13	29					
J41	CBAD/WRDBT12	30					
J42	CBAD/WRDBT11	32	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J43	CBAD/WRDBT10	33	6.0	0.0	0.5	V	1-4
J44	PRGM CNTR 15	34	1.6	0.0	0.5	V	1-4
J45	PRGM CNTR 15	34					
J46	PRGM CNTR 14	35					
J47	PRGM CNTR 14	35					
J48	PRGM CNTR 13	36					
J49	PRGM CNTR 13	36					
J50	PRGM CNTR 12	37					
J51	PRGM CNTR 12	37					
J52	PRGM CNTR 11	38					
J53	PRGM CNTR 11	38					
J54	PRGM CNTR 10	39					
J55	PRGM CNTR 10	39					
J56	PRGM CNTR 09	41					
J57	PRGM CNTR 09	41					
J58	PRGM CNTR 08	42					
J59	PRGM CNTR 08	42	1.6	0.0	0.5	V	1-4
J60	CBAD/WRDBT18	43	6.0	0.0	0.5	V	1-4
J61	CBAD/WRDBT19	44					
J62	CBAD/WRDBT20	45					
J63	CBAD/WRDBT21	46	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J64	CBAD/WRDBT08	47	6.0	0.0	0.5	V	1-4
J65	CBAD/WRDBT09	48					
J66	NDRO SELECT	49	6.0	0.0	0.5	V	1-4
J67	SR1 14 (GSS)	61	1.6	0.0	0.5	V	1-4
J68	LD P HISTORY	64	6.0	0.0	0.5	V	1-4
J69	FRCE I FETCH	65	1.6	0.0	0.5	V	1-4
J70	LDSHDWPRGCTR	66	6.0	0.0	0.5	V	1-4
J71	SR1 00 DMAEN	69	1.6	0.0	0.5	V	1-4
J72	SR1 01 CLIEN	71	6.0	0.0	0.5	V	1-4
J73	SR1 02 CIIEN	72					
J74	SR1 03 CIIIE	73	6.0	0.0	0.5	V	1-4
J75	PRGM CNTR 07	74	1.6	0.0	0.5	V	1-4
J76	PRGM CNTR 07	74					
J77	PRGM CNTR 06	76					
J78	PRGM CNTR 06	76					
J79	PRGM CNTR 05	77					
J80	PRGM CNTR 05	77					
J81	PRGM CNTR 04	78					
J82	PRGM CNTR 04	78					
J83	PRGM CNTR 03	79					
J84	PRGM CNTR 03	79	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J85	PRGM CNTR 02	80	1.6	0.0	0.5	V	1-4
J86	PRGM CNTR 02	80					
J87	PRGM CNTR 01	82					
J88	PRGM CNTR 01	82					
J89	PRGM CNTR 00	83					
J90	PRGM CNTR 00	83	1.6	0.0	0.5	V	1-4
J91	MDRO FULL	87	6.0	0.0	0.5	V	1-4
J92	KILL PIPE	89	1.6	0.0	0.5	V	1-4
J93	CBAD/WRDBT00	93	6.0	0.0	0.5	V	1-4
J94	CBAD/WRDBT01	94					
J95	CBAD/WRDBT02	95					
J96	CBAD/WRDBT03	96					
J97	CBAD/WRDBT04	97					
J98	CBAD/WRDBT05	98					
J99	CBAD/WRDBT06	99					
J100	CBAD/WRDBT07	100	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J100
Power supply current limit	3.20 A	NA	J1-J100
Forcing current, (Iol)	1.6 mA	± 160 uA	J1-J35
	6.0 mA	± 600 uA	J36-J43
	1.6 mA	± 160 uA	J44-J59
	6.0 mA	± 600 uA	J60-J66
	1.6 mA	± 160 uA	J67
	6.0 mA	± 600 uA	J68
	1.6 mA	± 160 uA	J69
	6.0 mA	± 600 uA	J70
	1.6 mA	± 160 uA	J71
	6.0 mA	± 600 uA	J72-J74
	1.6 mA	± 160 uA	J75-J90
	6.0 mA	± 600 uA	J91
	1.6 mA	± 160 uA	J92
	6.0 mA	± 600 uA	J93-J100
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	J1-J100
low, (VL)	0.0 V	± 100 mV	J1-J100
Output voltage measurement	NA	± 25 mV	J1-J100

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	CBAD/WRDBT15	27	2.4	-0.05	0.05	mA	1-3
K2	CBAD/WRDBT14	28					
K3	CBAD/WRDBT13	29					
K4	CBAD/WRDBT12	30					
K5	CBAD/WRDBT11	32					
K6	CBAD/WRDBT10	33					
K7	CBAD/WRDBT08	47					
K8	CBAD/WRDBT09	48					
K9	NDRO SELECT	49					
K10	CBAD/WRDBT00	93					
K11	CBAD/WRDBT01	94					
K12	CBAD/WRDBT02	95					
K13	CBAD/WRDBT03	96					
K14	CBAD/WRDBT04	97					
K15	CBAD/WRDBT05	98					
K16	CBAD/WRDBT06	99					
K17	CBAD/WRDBT07	100	2.4	-0.05	0.05	mA	1-3
Low level							
K18	CBAD/WRDBT15	27	0.4	-0.05	0.05	mA	1-3
K19	CBAD/WRDBT14	28	0.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K20	CBAD/WRDBT13	29	0.4	-0.05	0.05	mA	1-3
K21	CBAD/WRDBT12	30					
K22	CBAD/WRDBT11	32					
K23	CBAD/WRDBT10	33					
K24	CBAD/WRDBT08	47					
K25	CBAD/WRDBT09	48					
K26	NDRO SELECT	49					
K27	CBAD/WRDBT00	93					
K28	CBAD/WRDBT01	94					
K29	CBAD/WRDBT02	95					
K30	CBAD/WRDBT03	96					
K31	CBAD/WRDBT04	97					
K32	CBAD/WRDBT05	98					
K33	CBAD/WRDBT06	99					
K34	CBAD/WRDBT07	100	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K34
Power supply current limit	3.20 A	NA	K1-K34
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K17 K18-K34
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K34 K1-K34
Output current measurement	NA	± 5 uA	K1-K34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	MICRO IR 00	2	-18.0	-1.5	0.0	V	1-3
L2	MICRO IR 01	3	-18.0	-1.2	0.0	V	1-3
L3	MICRO IR 02	4	-18.0	-1.5	0.0	V	1-3
L4	DEST CLK HLD	5	-18.0	-1.2	0.0	V	1-3
L5	NDRO BIT MDE	21	-18.0	-1.5	0.0	V	1-3
L6	LDPGRG 12-15	22	-18.0	-1.2	0.0	V	1-3
L7	BOOT PRESENT	50	-18.0	-1.5	0.0	V	1-3
L8	JMP CND TRUE	52					
L9	MCRO 28 (D4)	53					
L10	MCRO 27 (D3)	54					
L11	MCRO 26 (D2)	55					
L12	MCRO 25 (D1)	56					
L13	MCRO 24 (D0)	57	-18.0	-1.5	0.0	V	1-3
L14	CB CLOCK	58	-18.0	-1.2	0.0	V	1-3
L15	MCRO 23(SA2)	59					
L16	MICRO IR 10	62					
L17	MICRO IR 05	67					
L18	CP CLOCK	68					
L19	2ND RNK FULL	70					
L20	ENBADRSGENSC	84	-18.0	-1.2	0.0	V	1-3
L21	CBMCRCNTFMT	85	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	ENBL WR DATA	86	-18.0	-1.5	0.0	V	1-3
L23	MCRO 16(SBO)	88	-18.0	-1.2	0.0	V	1-3
L24	INCR PC/SPC	91	-18.0	-1.5	0.0	V	1-3
L25	ENBCBAD/WRDA	92	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	OFF	NA	L1-L25
Power supply current limit	NA	NA	L1-L25
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L25
Pattern voltage high, (VH)	NA	NA	L1-L25
low, (VL)	NA	NA	L1-L25
Input voltage measurement	NA	± 75 mV	L1
	NA	± 60 mV	L2
	NA	± 75 mV	L3
	NA	± 60 mV	L4
	NA	± 75 mV	L5
	NA	± 60 mV	L6

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage measurement	NA	± 75 mV	L7-L13
	NA	± 60 mV	L14-L20
	NA	± 75 mV	L21,L22
	NA	± 60 mV	L23
	NA	± 75 mV	L24
	NA	± 60 mV	L25

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.9	7.4	uF	1-4
M2	PSFC	51	2.3	4.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 175 nF ± 105 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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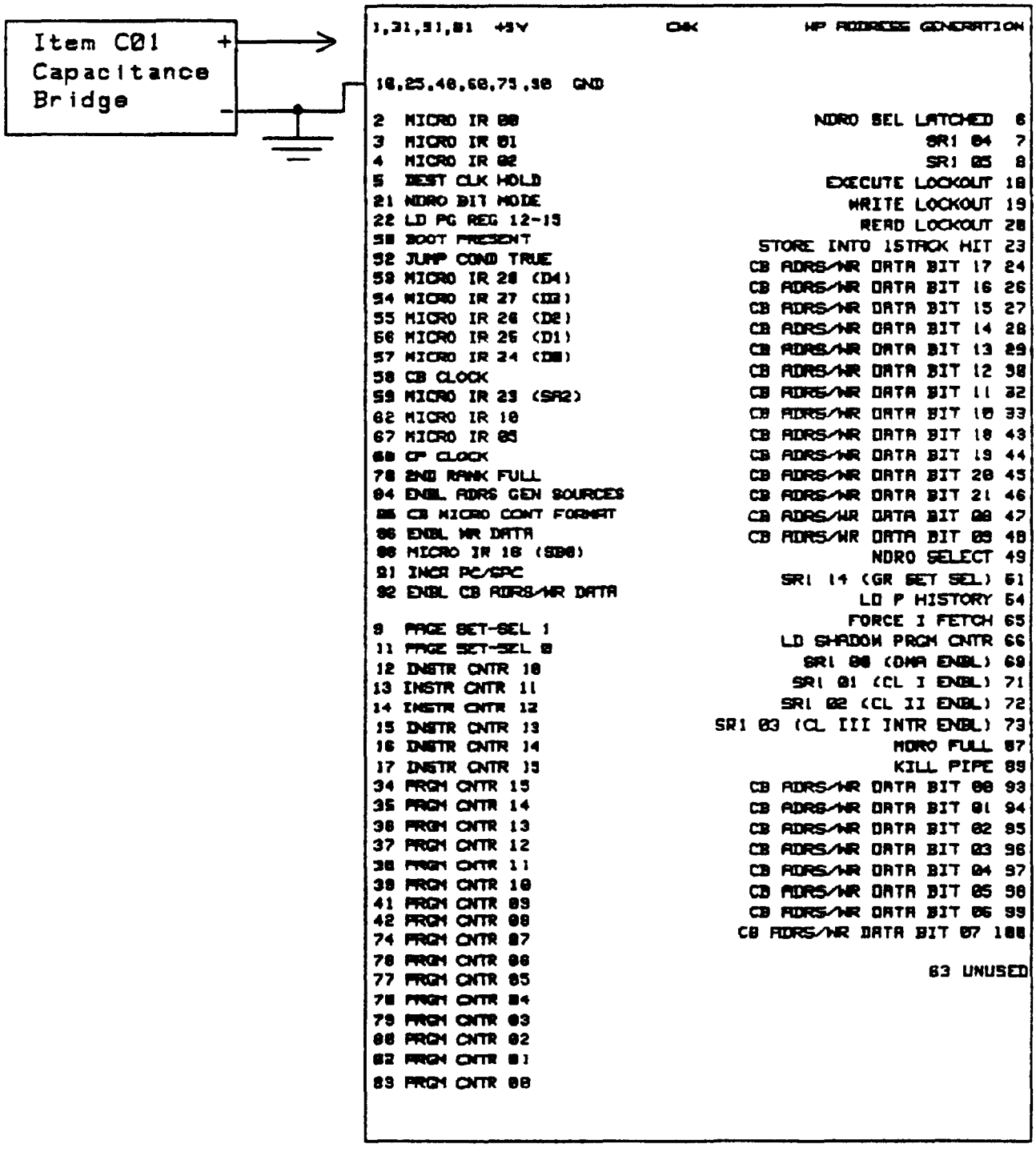


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	892	2139	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.20 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 62.4 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	3.20 A 3.20 A	P1 P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 NA NA	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 NA NA NA	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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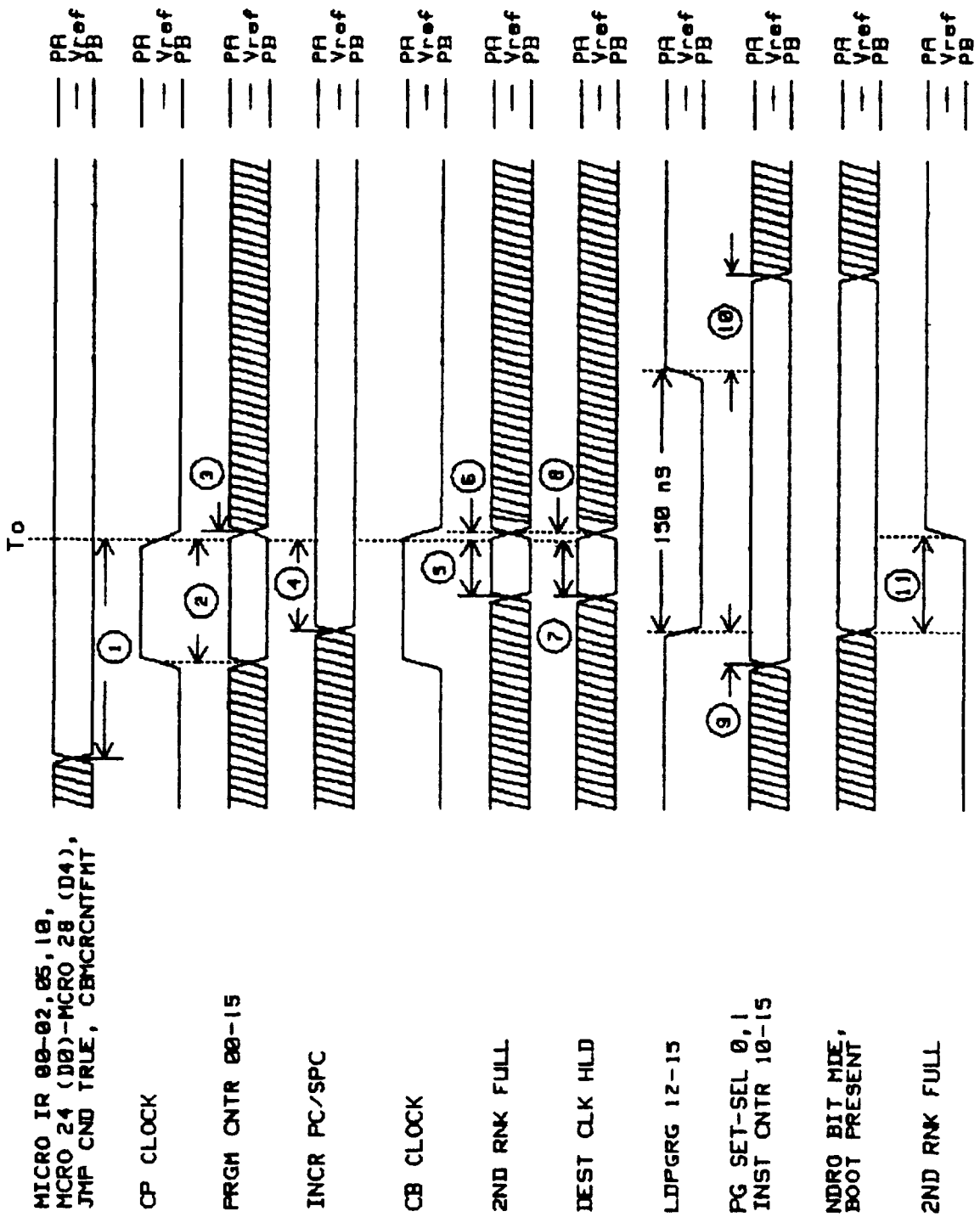


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) MICRO IR 00,MICRO IR 01,MICRO IR 02,MICRO IR 05, MICRO IR 10,MCRO 28 (D4), MCRO 27 (D3), MCRO 26 (D2), MCR0 25 (D1), MCRO 24 (D0), JMP CND TRUE, CBMCRCNTFMT to CP CLOCK. Setup time: 100 nS minimum.

2. (2) PRGM CNTR 15,PRGM CNTR 14,PRGM CNTR 13,PRGM CNTR 12, PRGM CNTR 11,PRGM CNTR 10,PRGM CNTR 09,PRGM CNTR 08,PRGM CNTR 07,PRGM CNTR 06,PRGM CNTR 05,PRGM CNTR 04,PRGM CNTR 03,PRGM CNTR 02,PRGM CNTR 01,PRGM CNTR 00 TO CP CLOCK. Setup time: 50 nS minimum.

3. (3) CP CLOCK to PRGM CNTR 15,PRGM CNTR 14,PRGM CNTR 13, PRGM CNTR 12,PRGM CNTR 11,PRGM CNTR 10,PRGM CNTR 09,PRGM CNTR 08,PRGM CNTR 07,PRGM CNTR 06,PRGM CNTR 05,PRGM CNTR 04, PRGM CNTR 03,PRGM CNTR 02,PRGM CNTR 01,PRGM CNTR 00. Hold time: 15 nS minimum.

4. (4) INCR PC/SPC to CP CLOCK. Set up time: 40 nS minimum.

5. (5) 2ND RNK FULL to CB CLOCK. Setup time: 15 nS minimum.

6. (6) CB CLOCK to 2ND RNK FULL. Hold time: 5 nS minimum.

7. (7) DEST CLK HLD to CP CLOCK. Setup time: 15 nS minimum.

8. (8) CP CLOCK to DEST CLK HLD. Hold time: 5 nS minimum.

9. (9) LDPGRG 12-15 to PG SET-SEL 1,PG SET-SEL 0,INST CNTR 10,INST CNTR 11,INST CNTR 12,INST CNTR 13,INST CNTR 14, INST CNTR 15. Setup time: 15 nS minimum.

10. (10) PG SET-SEL 1,PG SET-SEL 0,INST CNTR 10,INST CNTR 11, INST CNTR 12,INST CNTR 13,INST CNTR 14,INST CNTR 15 to LDPGRG 12-15. Hold time: 20 nS minimum.

11. (11) NDRO BIT MDE, BOOT PRESENT to 2ND RNK FULL. Hold time: 60 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	3.20 A 3.20 A	Q1,Q3 Q2,Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 NA NA	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test Condition
Output voltage verification				
high, (Voh)	NA	± 100 mV	NA	Q1,Q3
high, (Voh)	NA	± 100 mV	NA	Q2,Q4
low, (Vol)	NA	± 100 mV	NA	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

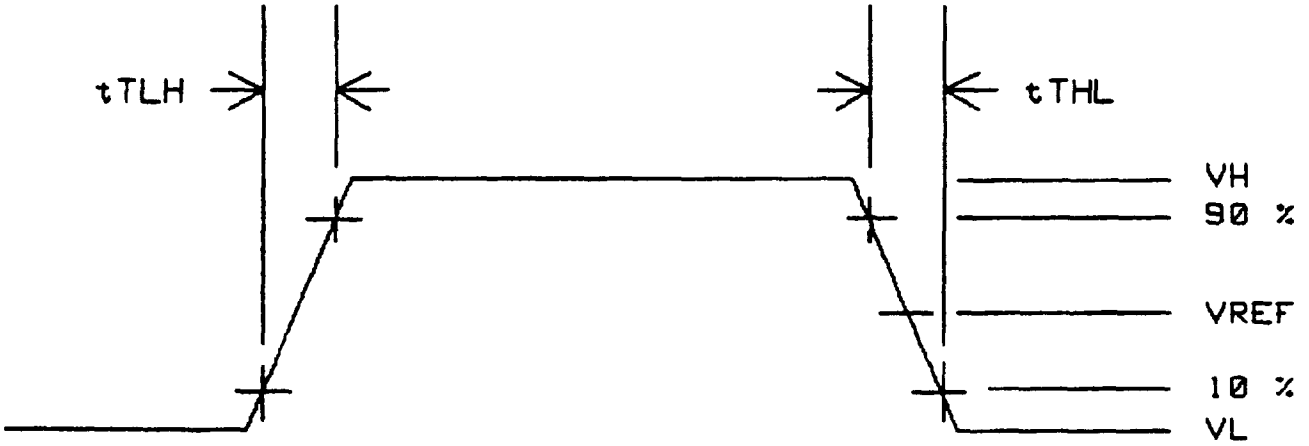
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CP CLOCK to MDRO FULL	-	-	-	30	30	30	nS	12
2	tpd	CP CLOCK to SR1 00 DMAEN	-	-	-	40	40	40	nS	12
2	tpd	CP CLOCK to SR1 04	-	-	-	40	40	40	nS	12
2	tpd	CP CLOCK to SR1 05	-	-	-	40	40	40	nS	12
2	tpd	CP CLOCK to SR1 14 (GSS)	-	-	-	40	40	40	nS	12
3	tpd	CP CLOCK to SR1 01 CLIEN	-	-	-	45	45	45	nS	12
3	tpd	CP CLOCK to SR1 02 CIIEN	-	-	-	45	45	45	nS	12
3	tpd	CP CLOCK to SR1 03 CIIIE	-	-	-	45	45	45	nS	12
4	tpd	MICRO IR 00 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 00 to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 01 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 01 to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 02 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 02 to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 10 to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MICRO IR 10 to KILL PIPE	-	-	-	55	55	55	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	MCRO 28 (D4) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 28 (D4) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 27 (D3) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 27 (D3) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 26 (D3) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 26 (D3) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 25 (D1) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 25 (D1) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	MCRO 24 (D0) to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	MCRO 24 (D0) to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	JMP CND TRUE to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	JMP CND TRUE to KILL PIPE	-	-	-	55	55	55	nS	12
4	tpd	CBMCRCNTFMT to FRCE I FETCH	-	-	-	55	55	55	nS	12
4	tpd	CBMCRCNTFMT to KILL PIPE	-	-	-	55	55	55	nS	12
5	tpd	MICRO IR 00 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	MICRO IR 00 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 01 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 01 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 02 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 02 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 10 to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MICRO IR 10 to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 28 (D4) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 28 (D4) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 27 (D3) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 27 (D3) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 26 (D2) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 26 (D2) to LD P HISTORY	-	-	-	60	60	60	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	MCRO 25 (D1) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 25 (D1) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	MCRO 24 (D0) to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	MCRO 24 (D0) to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	JMP CND TRUE to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	JMP CND TRUE to LD P HISTORY	-	-	-	60	60	60	nS	12
5	tpd	CBMCRCNTFMT to LDSHDWPRGCTR	-	-	-	60	60	60	nS	12
5	tpd	CBMCRCNTFMT to LD P HISTORY	-	-	-	60	60	60	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 15	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 14	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 13	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 12	-	-	-	80	80	80	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	ENBADRSGENSC to PRGM CNTR 11	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 10	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 09	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 08	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 07	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 06	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 05	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 04	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 03	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 02	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 01	-	-	-	80	80	80	nS	12
6	tpd	ENBADRSGENSC to PRGM CNTR 00	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 15	-	-	-	80	80	80	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MCRO 16(SB0) to PRGM CNTR 14	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 13	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 12	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 11	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 10	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 09	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 08	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 07	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 06	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 05	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 04	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 03	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 02	-	-	-	80	80	80	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MCRO 16(SB0) to PRGM CNTR 01	-	-	-	80	80	80	nS	12
6	tpd	MCRO 16(SB0) to PRGM CNTR 00	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 15	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 14	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 13	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 12	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 11	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 10	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 09	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 08	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 07	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 06	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 05	-	-	-	80	80	80	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	MCRO 23(SA2) to PRGM CNTR 04	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 03	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 02	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 01	-	-	-	80	80	80	nS	12
6	tpd	MCRO 23(SA2) to PRGM CNTR 00	-	-	-	80	80	80	nS	12
7	tpd	2ND RNK FULL to EXEC LCKOUT	-	-	-	30	30	30	nS	12
7	tpd	2ND RNK FULL to WRITE LCKOUT	-	-	-	30	30	30	nS	12
7	tpd	2ND RNK FULL to READ LCKOUT	-	-	-	30	30	30	nS	12
7	tpd	2ND RNK FULL to NDRO SEL LTC	-	-	-	30	30	30	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT17	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT16	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT15	-	-	-	20	20	20	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ENCBAD/WRDA to CBAD/WRDBT14	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT13	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT12	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT11	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT10	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT18	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT19	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT20	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT21	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT08	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT09	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT00	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT01	-	-	-	20	20	20	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ENCBAD/WRDA to CBAD/WRDBT02	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT03	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT04	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT05	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT06	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to CBAD/WRDBT07	-	-	-	20	20	20	nS	12
8	tpd	ENCBAD/WRDA to NDRO SELECT	-	-	-	20	20	20	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT15	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT14	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT13	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT12	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT11	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT10	-	-	-	65	65	65	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	ENBL WR DATA to CBAD/WRDBT08	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT09	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT00	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT01	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT02	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT03	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT04	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT05	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT06	-	-	-	65	65	65	nS	12
9	tpd	ENBL WR DATA to CBAD/WRDBT07	-	-	-	65	65	65	nS	12
10	tsu	MICRO IR 00 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MICRO IR 01 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MICRO IR 02 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MICRO IR 05 to CP CLOCK	100	100	100	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tsu	MICRO IR 10 to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 28 (D4) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 27 (D3) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 26 (D2) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 25 (D1) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	MCRO 24 (D0) to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	JMP CND TRUE to CP CLOCK	100	100	100	-	-	-	nS	12
10	tsu	CBMCRCNTFMT to CP CLOCK	100	100	100	-	-	-	nS	12
11	tsu	PRGM CNTR 15 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 14 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 13 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 12 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 11 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 10 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 09 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 08 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 07 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 06 to CP CLOCK	50	50	50	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tsu	PRGM CNTR 05 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 04 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 03 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 02 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 01 to CP CLOCK	50	50	50	-	-	-	nS	12
11	tsu	PRGM CNTR 00 to CP CLOCK	50	50	50	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 15	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 14	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 13	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 12	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 11	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 10	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 09	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 08	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 07	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 06	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 05	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 04	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 03	15	15	15	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	th	CP CLOCK to PRGM CNTR 02	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 01	15	15	15	-	-	-	nS	12
12	th	CP CLOCK to PRGM CNTR 00	15	15	15	-	-	-	nS	12
13	tsu	INCR PC/SPC to CP CLOCK	40	40	40	-	-	-	nS	12
14	tsu	2ND RNK FULL to CB CLOCK	15	15	15	-	-	-	nS	12
15	th	CB CLOCK to 2ND RNK FULL	5	5	5	-	-	-	nS	12
16	tsu	DEST CLK HLD to CP CLOCK	15	15	15	-	-	-	nS	12
17	th	CP CLOCK to DEST CLK HLD	5	5	5	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to PG SET-SEL 1	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to PG SET-SEL 0	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 10	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 11	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 12	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 13	15	15	15	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
18	tsu	LDPGRG 12-15 to INST CNTR 14	15	15	15	-	-	-	nS	12
18	tsu	LDPGRG 12-15 to INST CNTR 15	15	15	15	-	-	-	nS	12
19	th	PG SET-SEL 1 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	PG SET-SEL 0 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 10 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 11 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 12 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 13 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 14 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
19	th	INST CNTR 15 to LDPGRG 12-15	20	20	20	-	-	-	nS	12
20	tsu	NDRO BIT MDE to 2ND RNK FULL	60	60	60	-	-	-	nS	12
20	tsu	BOOT PRESENT to 2ND RNK FULL	60	60	60	-	-	-	nS	12

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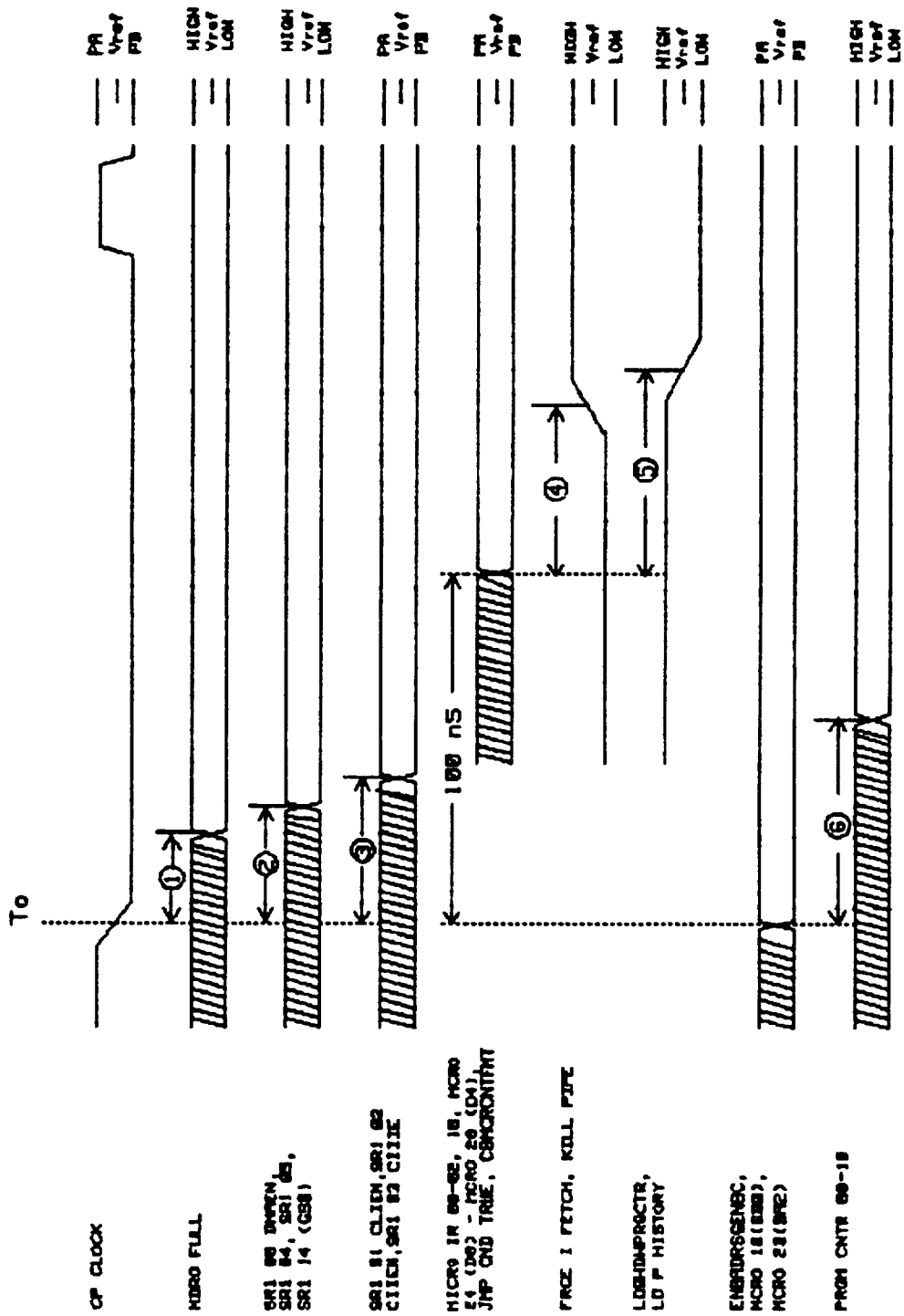


FIGURE 12a. Dynamic waveforms for table XVII.

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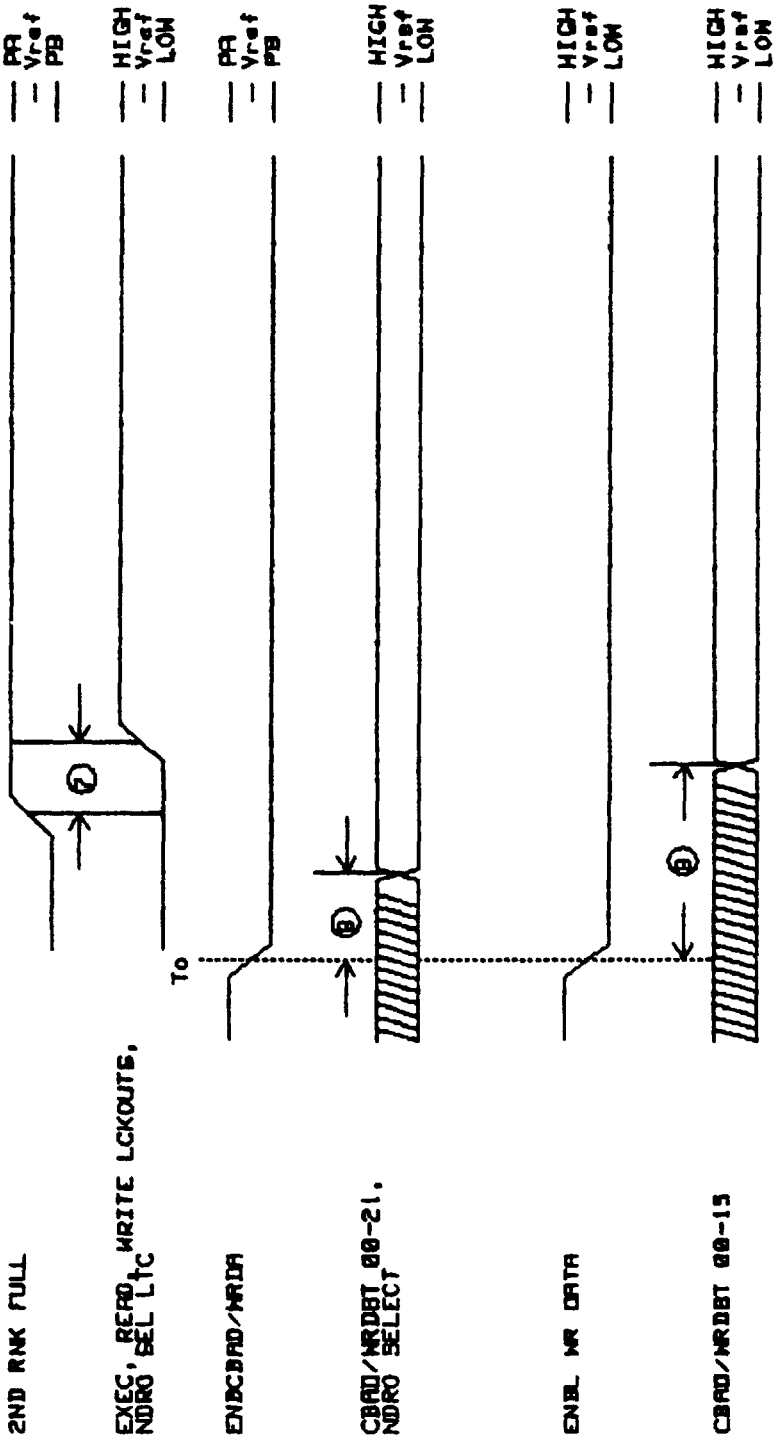


FIGURE 12b. Dynamic waveforms for table XVII.

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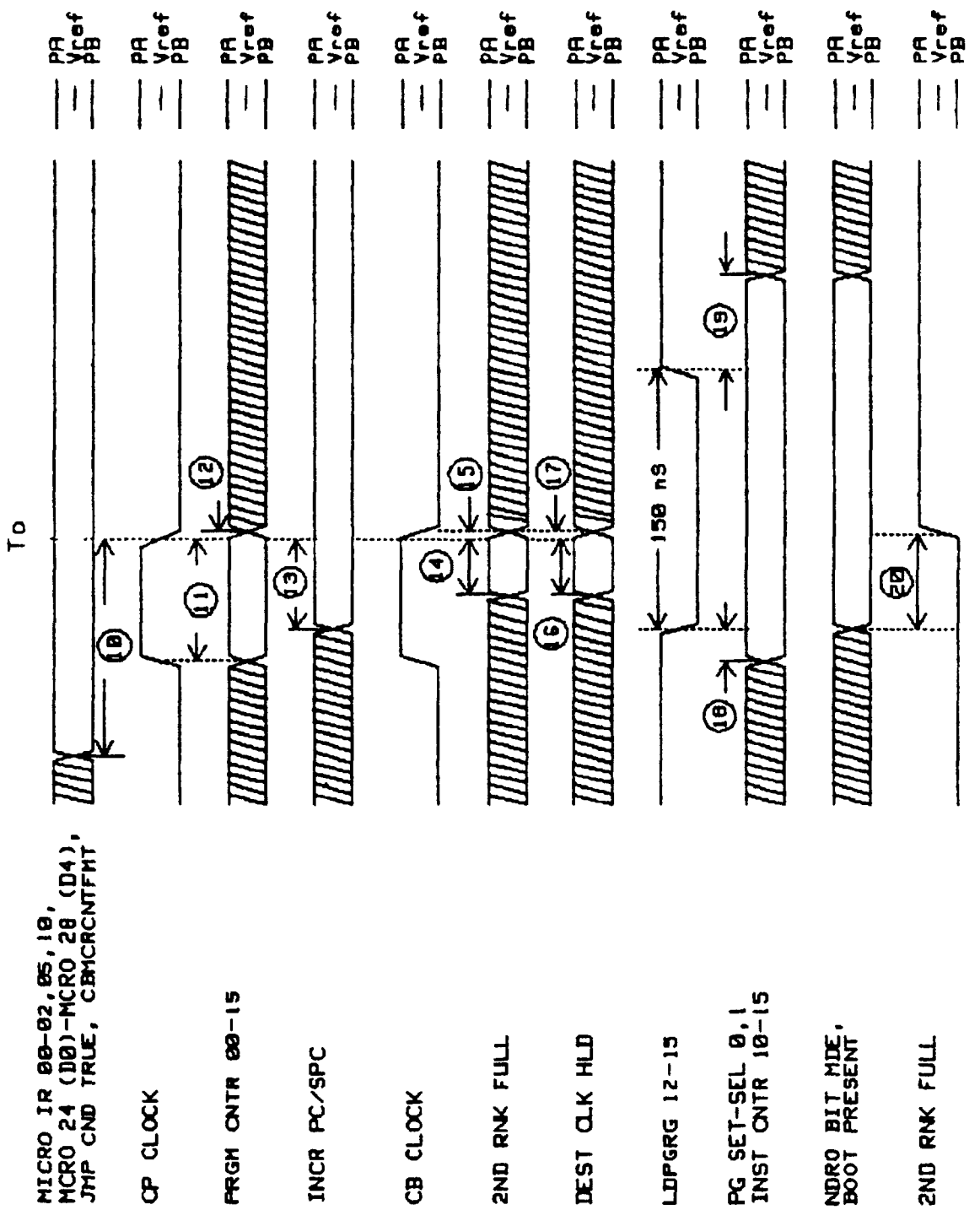


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	MICRO IR 00	2	2	75	pF	1-4
R2	MICRO IR 01	3				
R3	MICRO IR 02	4	2	75	pF	1-4
R4	DEST CLK HLD	5	2	100	pF	1-4
R5	NDRO BIT MDE	21	2	75	pF	1-4
R6	LDPGRG 12-15	22	4	100	pF	1-4
R7	BOOT PRESENT	50	2	125	pF	1-4
R8	JMP CND TRUE	52	2	75	pF	1-4
R9	MCRO 28 (D4)	53	6	100	pF	1-4
R10	MCRO 27 (D3)	54	6	75	pF	1-4
R11	MCRO 26 (D2)	55				
R12	MCRO 25 (D1)	56	6	75	pF	1-4
R13	MCRO 24 (D0)	57	2	175	pF	1-4
R14	CB CLOCK	58	2	50	pF	1-4
R15	MCRO 23(SA2)	59	4	150	pF	1-4
R16	MICRO IR 10	62	10	125	pF	1-4
R17	MICRO IR 05	67	2	75	pF	1-4
R18	CP CLOCK	68	18	175	pF	1-4
R19	2ND RNK FULL	70	6	175	pF	1-4
R20	ENBADRSGENSC	84	4	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CBMCRCNT FMT	85	4	125	pF	1-4
R22	ENBL WR DATA	86	4	150	pF	1-4
R23	MCRO 16(SBO)	88	6	100	pF	1-4
R24	INCR PC/SPC	91	4	125	pF	1-4
R25	ENBCBAD/WRDA	92	8	150	pF	1-4
Outputs						
R26	CBAD/WRDBT17	24	4	75	pF	1-4
R27	CBAD/WRDBT16	26	4	75	pF	1-4
R28	CBAD/WRDBT15	27	2	75	pF	1-4
R29	CBAD/WRDBT14	28				
R30	CBAD/WRDBT13	29				
R31	CBAD/WRDBT12	30				
R32	CBAD/WRDBT11	32				
R33	CBAD/WRDBT10	33	2	75	pF	1-4
R34	CBAD/WRDBT18	43	4	75	pF	1-4
R35	CBAD/WRDBT19	44				
R36	CBAD/WRDBT20	45				
R37	CBAD/WRDBT21	46	4	75	pF	1-4
R38	CBAD/WRDBT08	47	2	75	pF	1-4
R39	CBAD/WRDBT09	48				
R40	CBAD/WRDBT00	93	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	CBAD/WRDBT01	94	2	75	pF	1-4
R42	CBAD/WRDBT02	95				
R43	CBAD/WRDBT03	96				
R44	CBAD/WRDBT04	97				
R45	CBAD/WRDBT05	98	2	75	pF	1-4
R46	CBAD/WRDBT06	99	2	50	pF	1-4
R47	CBAD/WRDBT07	100	2	75	pF	1-4
Bidirectionals						
R48	PRGM CNTR 15	34	6	100	pF	1-4
R49	PRGM CNTR 14	35				
R50	PRGM CNTR 13	36				
R51	PRGM CNTR 12	37				
R52	PRGM CNTR 11	38				
R53	PRGM CNTR 10	39				
R54	PRGM CNTR 09	41				
R55	PRGM CNTR 08	42				
R56	PRGM CNTR 07	74				
R57	PRGM CNTR 05	77				
R58	PRGM CNTR 04	78				
R59	PRGM CNTR 03	79				
R60	PRGM CNTR 02	80	6	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	PRGM CNTR 01	82	6	100	pF	1-4
R62	PRGM CNTR 00	83	6	125	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R62
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R62
Capacitance measurement	NA	± 3.65 pF	R1-R3
	NA	± 4.9 pF	R4
	NA	± 3.65 pF	R5
	NA	± 4.8 pF	R6
	NA	± 6.15 pF	R7
	NA	± 3.65 pF	R8
	NA	± 4.7 pF	R9
	NA	± 3.45 pF	R10-R12
	NA	± 8.65 pF	R13
	NA	± 2.4 pF	R14
	NA	± 7.3 pF	R15
	NA	± 5.75 pF	R16
	NA	± 3.65 pF	R17
	NA	± 7.85 pF	R18

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 8.45 pF	R19
	NA	± 4.8 pF	R20
	NA	± 6.05 pF	R21
	NA	± 7.3 pF	R22
	NA	± 4.7 pF	R23
	NA	± 6.05 pF	R24
	NA	± 7.1 pF	R25
	NA	± 3.55 pF	R26,R27
	NA	± 3.65 pF	R28-R33
	NA	± 3.55 pF	R34-R37
	NA	± 3.65 pF	R38-R45
	NA	± 2.4 pF	R46
	NA	± 3.65 pF	R47
	NA	± 4.7 pF	R48-R61
	NA	± 5.95 pF	R62

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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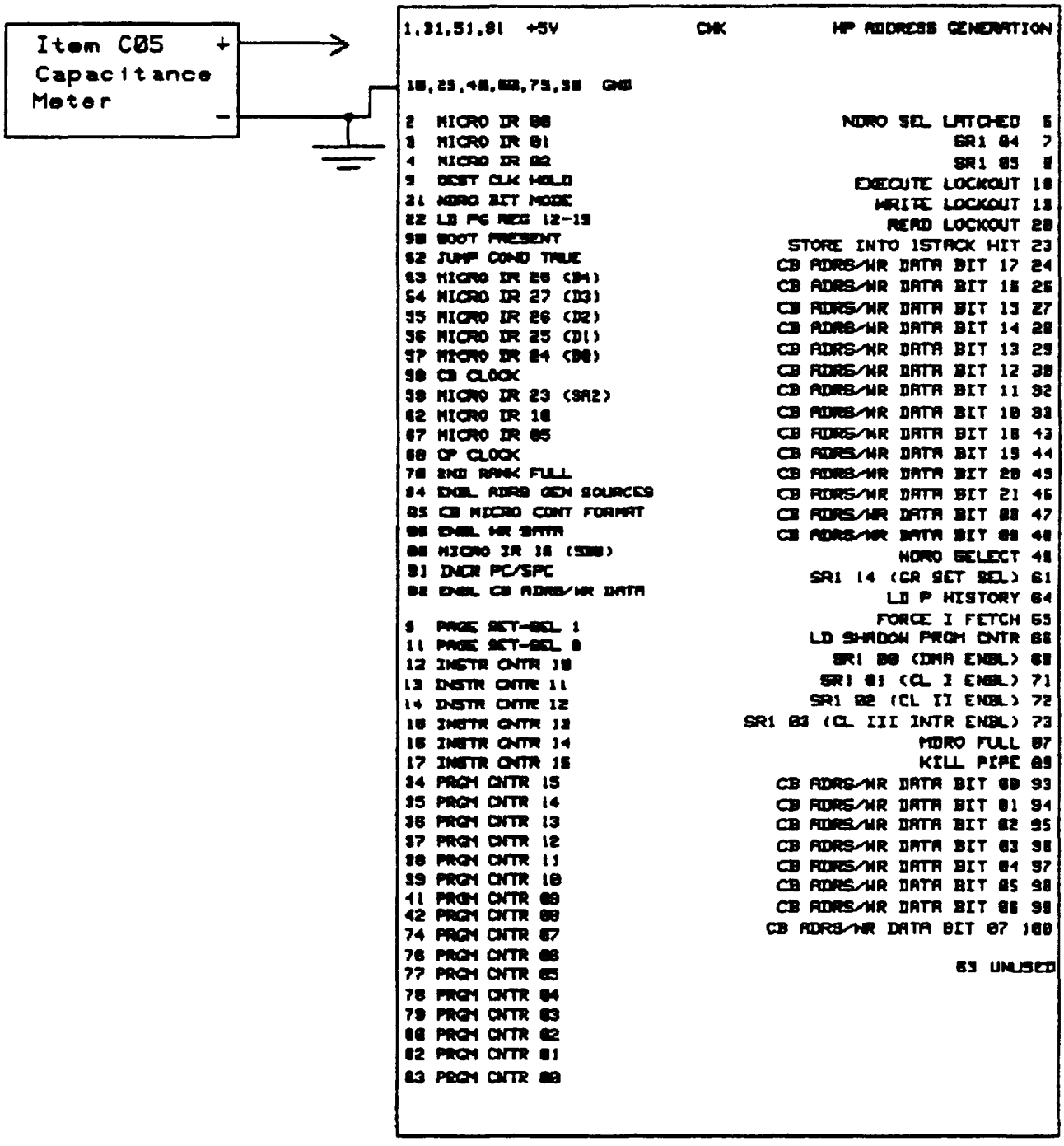


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	3.20 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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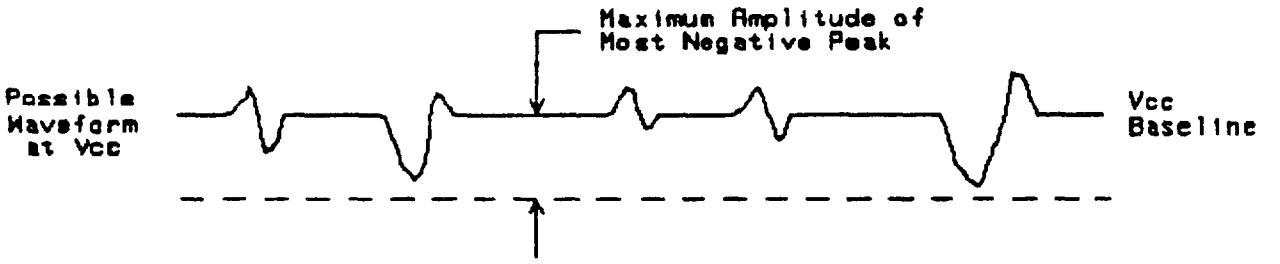
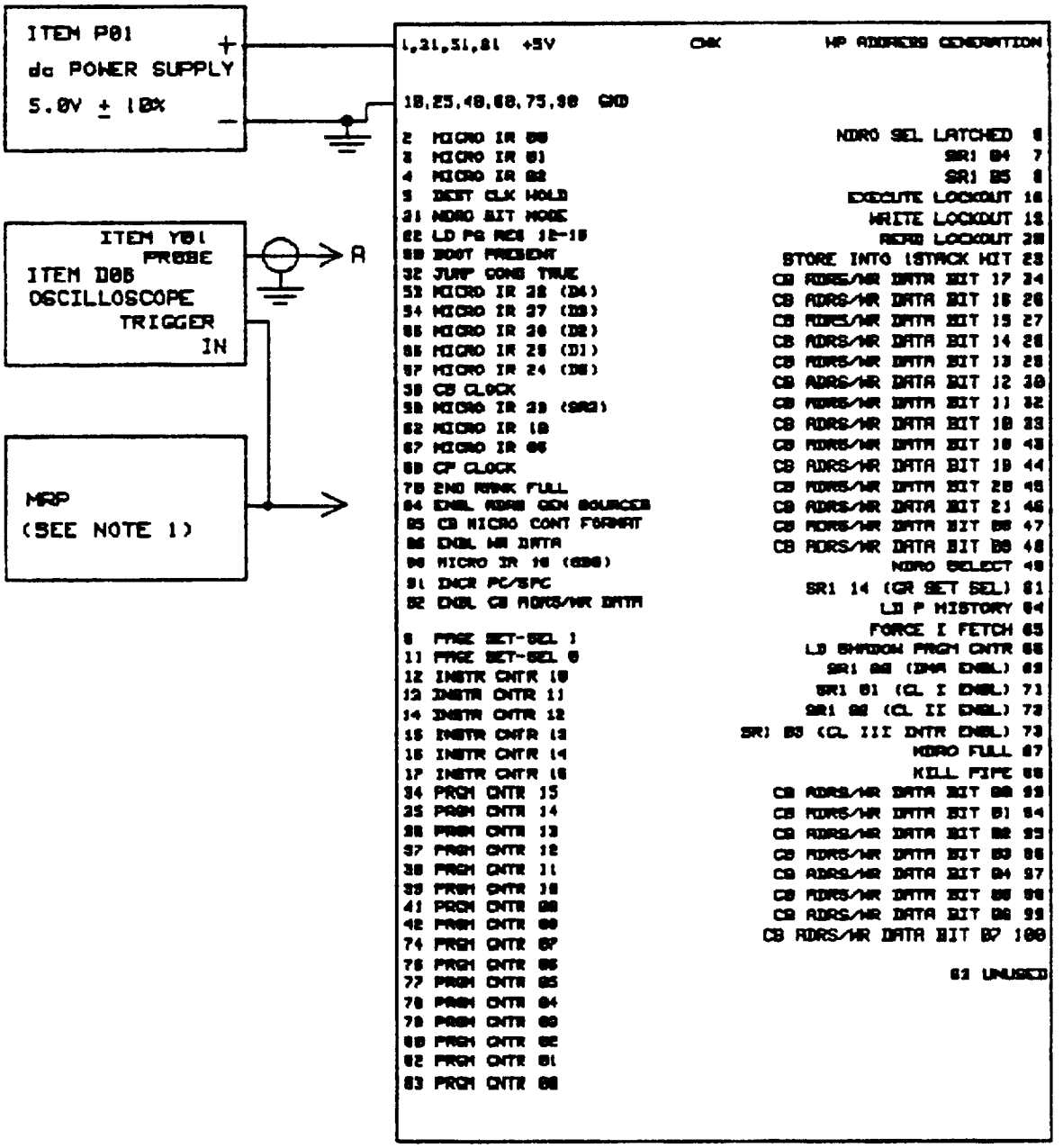


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

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a. Paragraph Number and Wording.

b. Recommended Wording:

c. Reason/Rationale for Recommendation

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